Grey Box: Reading Closed Agents Through Their APIs Alone

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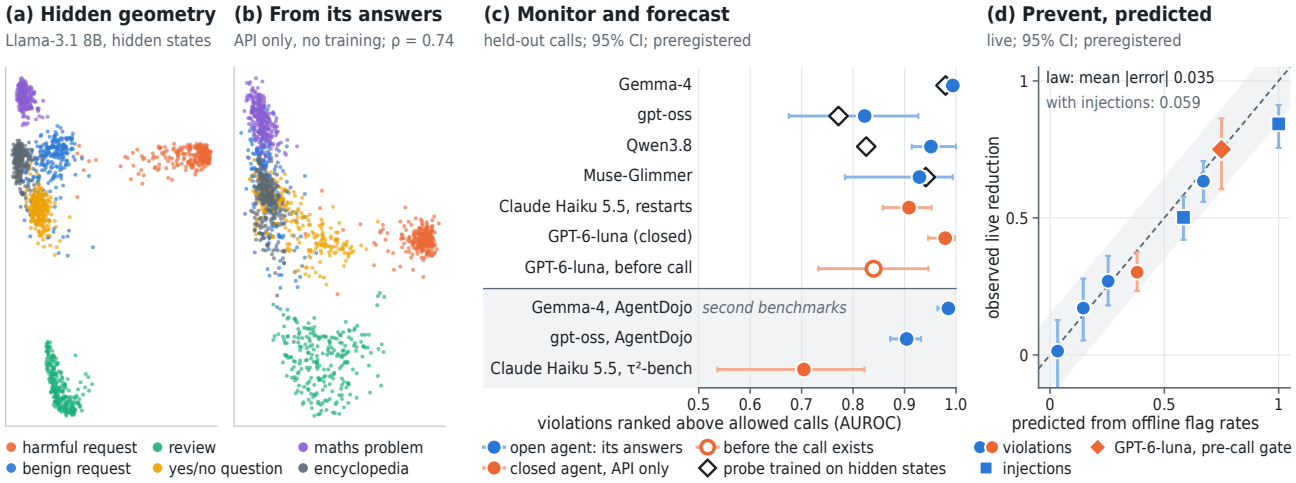


Figure 1. **Reading closed agents through their own APIs, with no training.** (a) An open model's hidden geometry on 1,440 held-out contexts. (b) The same contexts, placed only by the model's yes/no answers in exact Fisher–Rao coordinates (Spearman 0.74 between the distances; 0.41–0.74 over eight models). (c) A fixed read-out of an agent's answers to 8 policy questions ranks its violations within each kind of task, about as well as a probe trained on its hidden states. Orange: closed agents. Hollow: GPT-6-luna before its call exists (0.84 within task and tool, against 0.78 for position alone; 0.92 among calls at the same position). Below the line: second benchmarks. Claude Haiku 5.5's restarts are one kind of task; over all its actions it scores 0.475 (Table 2). (d) Live prevention against the prediction from offline flag rates (band ±0.15). Diamond: the pre-call gate on GPT-6-luna. 95% intervals; all preregistered.

Abstract

Probes on an agent's hidden states can stop an unsafe action before it runs and catch what monitors of its text miss (Fayyazi et al., 2026), but they need the weights, and the most capable agents are closed. We ask what a closed agent's own API reveals of its state, with no training and no other model, and what this buys for safety. An API cannot reveal hidden coordinates (a gauge symmetry), but it reveals geometry: yes/no questions asked in side branches give answer laws whose Fisher–Rao distances are exact Euclidean distances in arcsine coordinates, with a known noise. On eight open models this geometry tracks each model's hidden geometry better than the input or the replies do, and it is the geometry that models share. Three consequences follow, and we test each in preregistered experiments. *At a proposed call*, the transcript decides the violation, so any capable answerer reads it: a fixed read-out of the agent's own answers ranks violations at AUROC 0.82–0.99 on four open agents, close to a probe on their hidden states, 0.90–0.99 on prompt injection and 0.98 on GPT-6-luna, and another model's answers do as well. *Before the call*, only the agent's own policy can forecast it: GPT-6-luna's own answers rank its coming violation better than another model's answers in its seat (0.84 vs 0.73). *Live*, gates on these reads prevent violations by amounts that offline flag rates predict; a gate on GPT-6-luna's pre-call state cut its violating episodes from 32% to 8%, against 18% for a gate on position at the same review rate. Every test is preregistered, and we release the tool.

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1. Introduction

Interpretability reads what a model knows and what it will do from its hidden state (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025). For agents, a probe on the hidden state can stop an unsafe action before it runs (Fayyazi et al., 2026). These tools need the weights. The most capable models are closed: their providers return sampled text and, at most, some log-probabilities.

Why read the state, not the text. Agents make calls that they are not authorized to make, and when asked, they may say that the call is fine (Fayyazi et al., 2026). Chain-of-thought often leaves out what drives an answer (Turpin et al., 2023; Chen et al., 2025). A rule check that the operator writes catches only the rules that it names. In our data, asked directly whether its unapproved production restarts complied, Claude Haiku 5.5 mostly said yes (AUROC 0.32, below chance); its answers to eight specific questions about the same calls ranked them at 0.91.

What can an API reveal? Not hidden coordinates: any change of basis that the next layers undo leaves every output unchanged (Proposition 1). Geometry survives. We ask the model yes/no questions in *side branches*: copies of the conversation, each with one question appended, sent through its API and then discarded, so the agent never sees them. In the angles $\vartheta = 2 \arcsin \sqrt{P(\text{YES})}$, the Fisher–Rao distance between answer laws is Euclidean (Theorem 1), and its noise is known. Nothing is trained.

From what the API reveals to what it buys. Our results form a chain of three consequences of one finding.

(i) *The API reveals the shared geometry* (Section 3). On eight open models, the answer geometry tracks each model's hidden geometry better than the input text or the replies do. A model's answers match other models' hidden geometry exactly as well as its own (excess +0.002). Claude Haiku 5.5's answers agree with open models' hidden geometry at 0.45–0.51.

(ii) *At the call, the shared state suffices* (Sections 4.1 and 4.2). The transcript and the call decide a violation (Corollary 1). So a fixed read-out of the agent's answers, never fit to labels, ranks violations nearly as well as a probe on its hidden states, and another model's answers do as well. It carries to prompt injection and to two closed agents.

(iii) *Before the call, the forecast needs the agent* (Section 4.3). Before a call exists, the transcript does not decide it; only the agent's own policy does (Proposition 3). On GPT-6-luna, its own answers rank the coming violation better than another model's answers in its seat.

Prevention (Section 4.4). Gates on these reads prevent violations live, by amounts that offline flag rates predict. A gate on GPT-6-luna's pre-call state prevented more than a gate on position at the same review rate.

Contributions. (1) *Theory* (Section 2). Parts are classical (Fisher–Rao coordinates, computational invariance); new are an exact cross-fit estimator from sampled answers, a triangulation for closed models, a label-free ceiling on a read-out, and the split between monitoring at the call and forecasting before it. (2) *Evidence* on eight open models, four open agents, two closed agents and three benchmarks (Sections 3 and 4). (3) A released tool (Section 5).

Preregistration. We hash-locked every confirmatory test before its data existed, and published the later locks before the run. A result is preregistered unless marked exploratory. Section L lists the tests behind each claim; the full record of every test is public with the code.

2. Theory

2.1. Setting

A language model reads a context x (tokens) and reaches an internal state $s(x)$, a deterministic function of the tokens. An outside observer sees only outputs: sampled text and, on some APIs, the log-probabilities of the next token. A *battery* $Q = (q_1, \dots, q_m)$ of yes/no questions is asked in side branches: each question is appended to x in its own call, so the model's real conversation never sees it, and the answers are independent given x. The *interrogation state* is $\Psi_Q(x) = \big(P(\text{YES} \mid x \oplus q_j)\big)_j$, and the answers follow the product law $P_x = \bigotimes_j \text{Bernoulli}(\Psi_{q_j}(x))$. Where a model is open, its hidden state $h(x) \in \mathbb{R}^d$ at a layer is the ground truth that we test against.

2.2. What cannot be read: a gauge symmetry

Proposition 1 (gauge)**.** *Take a pre-norm residual network with RMS normalisation, with the normalisation gains folded into the adjacent weights. For every orthogonal G on the residual stream, the network whose embedding, write matrices and write biases are multiplied by G on the left, and whose read matrices and unembedding are multiplied by $G^\top$ on the right, has hidden states Gh and the same output distribution on every input. At a layer that is read only through linear maps, the same holds for every invertible G, with G^{-1} in place of $G^\top$.*

No estimator that uses only outputs can recover hidden *coordinates* beyond this group (computational invariance; Ashkboos et al., 2024). Distances and angles between states are invariant under the orthogonal group, so the *geometry* of the states is the natural target. Post-block normalisation (Gemma-3, OLMo-3) restricts G to the maps that commute with its gains (Section B).

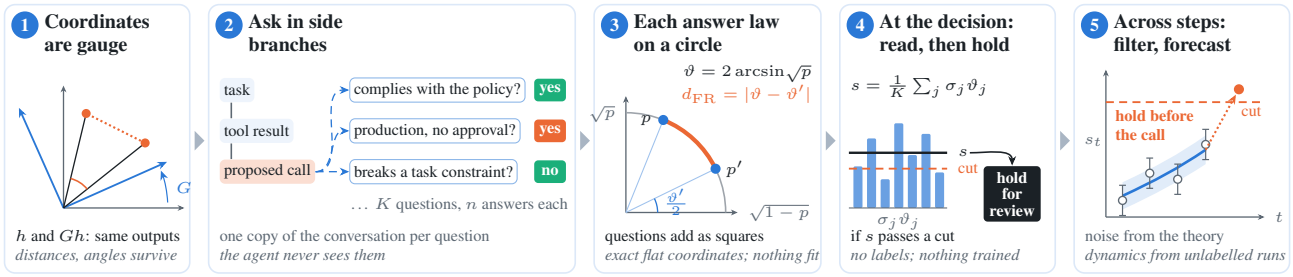


Figure 2. **The method, with no training and nothing but the API.** (1) Hidden coordinates are not observable (Proposition 1). (2) At a proposed call, the agent answers yes/no questions about the call in side branches of its own conversation, which the conversation never sees. (3) A Bernoulli law p is the point $(\sqrt{1-p}, \sqrt{p})$ on a quarter circle, and its Fisher–Rao distance to another is twice the arc between them; independent questions add as squares, so the angles are exact flat coordinates (Theorem 1). (4) The angles, signed by each question's wording, are averaged into a fixed score that holds the call for review before it runs. (5) Across an episode, the angles are noisy measurements (error bars: the theory's noise) of a state whose dynamics are identified from unlabelled runs (Proposition 5); a Kalman filter gives the estimated state and a forecast of the next one, which can hold the agent before it proposes the call. Asked about any context, the same coordinates give the observable geometry of Section 3.

2.3. What can be read: the observable geometry

Let $d_Q(x, x')$ be the Fisher–Rao geodesic distance between the answer laws P_x and $P_{x'}$.

Theorem 1 (observable geometry)**.** *(a)* Exact flat coordinates. *With* $\vartheta_j = 2\arcsin\sqrt{\Psi_{q_j}}$, $d_Q(x,x')^2 = \sum_j \big(\vartheta_j(x) - \vartheta_j(x')\big)^2$. *(b)* Invariance and uniqueness. d_Q *depends on the outputs only, so it is gauge invariant; up to a constant, the Fisher metric is the only Riemannian metric on the simplex over* $\{0,1\}^m$ *invariant under Markov morphisms (Čencov), and* d_Q *is its geodesic distance on the independent answer laws. (c)* Relation to the hidden state. *If* $\Psi_j = \sigma(a_j(s))$ *with smooth* a_j*, then for nearby states* $d_Q^2 = \Delta^\top G_Q(s)\,\Delta + O(|\Delta|^3)$*, with the pulled-back Fisher metric* $G_Q = \sum_j \Psi_j(1-\Psi_j)\nabla a_j \nabla a_j^\top$. *(d)* Estimation from sampled answers. *Let* $\mu_n(p) = \mathbb{E}\,\hat\vartheta$ *for the Anscombe angle* $\hat\vartheta = 2\arcsin\sqrt{(k+3/8)/(n+3/4)}$*,* $k \sim \mathrm{Bin}(n,p)$*;* μ_n *is known, strictly increasing, and* $\mu_n \to \vartheta$*. For two independent halves* A, B *of* n *samples each,* $\hat D^2(x,x') = \sum_j \big(\hat\vartheta_j^A(x) - \hat\vartheta_j^A(x')\big)\big(\hat\vartheta_j^B(x) - \hat\vartheta_j^B(x')\big)$ *satisfies* $\mathbb{E}\hat D^2 = \sum_j \big(\mu_n(\Psi_j(x)) - \mu_n(\Psi_j(x'))\big)^2$ *exactly, for every* Ψ*, when every cell has the same* n*.*

Parts (a)–(c) are classical information geometry (Amari, 2016); part (d) is ours. Part (a) is a change of variables: nothing is fit, and classical scaling of d_Q gives coordinates that are exact up to a rigid motion. Part (b) says these are the only natural coordinates; Picozzi (2026) study the same object for next-token laws. Part (c) ties them to the hidden state: agreement should rise as more questions give G_Q more rank. Part (d) makes the geometry measurable on any chat API, at $m \cdot n$ one-token calls per context (exactness needs the same n in every cell and answers that are yes or no; Section B).

Checking a closed model's reconstruction. A closed model's hidden states cannot be seen, so how well its answer geometry matches its own hidden geometry cannot be measured directly. It can be estimated, because hidden geometries share a common part across models (Section 3).

Proposition 2 (triangulation)**.** *On the vector of all context-pair distances, let the hidden geometry of model* m *be* $H_m = \lambda_m C + e_m$ *and the answer geometry of answerer* a *be* $A_a = \kappa_a C + f_a$ *(standardized; the parts* e_m*,* f_a *uncorrelated with* C *and across models). Then* $\rho(H_a, H_b) = \lambda_a\lambda_b$ *and* $\rho(A_a, H_b) = \kappa_a\lambda_b$ *for* $a \neq b$*, and the unseen agreement is* $\rho(A_c, H_c) = \kappa_c\lambda_c + \rho(f_c, e_c)$*. Open models identify the* λ_b*; a closed model's answers against them identify* κ_c*.*

With no own-model excess ($\rho(f_c, e_c) = 0$; tested in Section 3) and λ_c in the range of open models, $\kappa_c \cdot \mathrm{median}(\lambda)$ estimates the closed model's own agreement. Leaving each open model out, as if closed, tests the estimator against the truth.

Where an API returns full log-probabilities, the final state's geometry is recovered exactly, not estimated (Theorem 2, Section B, after Carlini et al. (2024, App. H)); closed APIs do not return them, so the grey box does not use this.

2.4. From the geometry to monitoring agents

Setting. At step t of an episode the operator holds the visible tokens V_t (instructions, task, earlier calls and their results) and sees the proposed call A_t; $Y \in \{0,1\}$ marks a policy violation. A monitor scores (V_t, A_t) and holds the call when the score passes a cut; its budget β is the share of calls it holds.

Corollary 1 (the transcript suffices)**.** *Let* $Y = g(A_t, V_t, W)$*, where the world facts* W *satisfy* $W \perp\!\!\!\perp Z_t^- \mid V_t$ *and* Z_t^- *are all tokens before the call, hidden ones included. Then* $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$*: no monitor of the agent's internals has a lower Bayes risk than the best monitor of*

(V_t, A_t).

So a monitor needs the transcript and the call, not the weights; the question is what to compute from them.

Proposition 3 (the forecast needs the agent)**.** *Before step t the agent acts by $A_t \sim \pi(\cdot \mid V_t, Z_t^-)$; let $\bar\pi(a \mid V_t)$ be this policy averaged over the hidden states consistent with V_t, and $r(a, V_t) = P(Y \mid a, V_t)$. The Bayes forecast is $f = \sum_a \bar\pi(a \mid V_t)\, r(a, V_t)$. A reader that forecasts with another policy π', $f' = \sum_a \pi'(a \mid V_t)\, r(a, V_t)$, has squared-loss excess risk*

$$\mathbb{E}(Y - f')^2 - \mathbb{E}(Y - f)^2 = \mathbb{E}\,\Delta^2 \le \mathbb{E}\,\|\bar\pi - \pi'\|_1^2,$$

with $\Delta = \sum_a (\bar\pi - \pi')(a \mid V_t)\, r(a, V_t)$. The excess is zero once the call A_t is visible.

Proof sketch. $f = \mathbb{E}[Y \mid V_t]$, so $\mathbb{E}(Y-f')^2 = \mathbb{E}(Y-f)^2 + \mathbb{E}(f - f')^2$; and $|f - f'| \le \|\bar\pi - \pi'\|_1$ since $0 \le r \le 1$. Given A_t, both forecasts equal $r(A_t, V_t)$. □

So at the call any capable reader can match the agent (Corollary 1); before it, a reader loses exactly as much as its policy differs from the agent's on violating actions. Another model's answers can forecast well only where it would act like the agent (tested in Section 4.3).

The observable state at the decision. At a proposed call we ask the agent, through its own API, K yes/no questions about the call in side branches (Figure 2). Their angles $\vartheta_t \in [0, \pi]^K$ are its observable state, in the coordinates of Theorem 1. The score is a fixed read-out, the mean angle in each question's violation direction:

$$s_t = \frac{1}{K} \sum_{j=1}^{K} \sigma_j\, \vartheta_{tj}, \qquad \sigma_j \in \{-1, +1\}. \tag{1}$$

Here $\sigma_j = -1$ for the question "does it comply" and $+1$ for the others. No label is used and nothing is fit. Self-policing is the one-question case: hold the call when the agent's own answers say that it does not comply.

Proposition 4 (a label-free ceiling)**.** *Let N be the sampling-noise variance of s_t and $\mathrm{SNR} = \mathrm{Var}(s_t)/N - 1$; neither uses a label. If the noise is independent of Y and s_t is Gaussian with equal variance in the two classes, then for a violation rate p, $\mathrm{AUROC}(s_t) \le \Phi\big(\sqrt{\mathrm{SNR}^+/(2p(1-p))}\big)$; at $\mathrm{SNR} \le 0$ the read-out cannot rank violations.*

Proof sketch. With class gap Δ and within-class variance $\sigma_w^2 \ge N$, $\mathrm{Var}(s_t) = \sigma_w^2 + p(1-p)\Delta^2$, so $p(1-p)\Delta^2 \le \mathrm{SNR} \cdot N \le \mathrm{SNR} \cdot \sigma_w^2$; and $\mathrm{AUROC} = \Phi(\Delta/\sqrt{2}\sigma_w)$. □

In practice we use the nominal noise $\frac{1}{K^2}\sum_j (m_j + \frac{1}{2})^{-1}$ for m_j parsed answers. It is exact at $P(\textsc{yes}) = \frac{1}{2}$ and overstates the noise near 0 or 1 (with 4 answers at $P = 0.01$ the exact variance is 0.013, not 0.222), so the nominal SNR is conservative: a low value flags a setting, and a high value supports it (Section 4.1). More answers lower N and raise the ceiling: 16 answers instead of 4 raised the pre-call ranking from 0.65 to 0.73.

The state across steps. Before each call we measure y_t, the angles of the agent's answers to K questions about its *next* action. By Theorem 1, $y_t = z_t + v_t$: the observable state plus sampling noise whose variance the number of answers fixes ($\approx 1/(n + \frac{1}{2})$ per question), independent across steps. We model $z_{t+1} = Fz_t + c + w_t$ and identify the model from unlabelled runs of the same agent.

Proposition 5 (identification through the noise)**.** *If v_t is independent across steps and of (z_s, w_s), and $\mathrm{Cov}(z_t, z_{t-1})$ is invertible, then $F = \mathrm{Cov}(y_{t+1}, y_{t-1})\,\mathrm{Cov}(y_t, y_{t-1})^{-1}$ and $\Sigma_z = F^{-1}\mathrm{Cov}(y_{t+1}, y_t)$, whatever the law of v_t. Regressing y_{t+1} on y_t instead shrinks F by the noise.*

Proof sketch. The noise at $t \pm 1$ is independent of everything at other steps, so $\mathrm{Cov}(y_{t+1}, y_{t-1}) = F\,\mathrm{Cov}(z_t, z_{t-1}) = F\,\mathrm{Cov}(y_t, y_{t-1})$ and $\mathrm{Cov}(y_{t+1}, y_t) = F\Sigma_z$. □

A Kalman filter (with the noise taken from the data: for nearly deterministic answers the nominal value overstates it about threefold) then gives the estimated state $\mathbb{E}[z_t \mid y_{\le t}]$, measured before the call is proposed, and the forecast $\mathbb{E}[z_t \mid y_{<t}]$ from earlier steps alone; the gap between the noise that Proposition 5 implies and the theory's noise tests the model.

The cut. The cut is set on the agent's own unlabelled traffic and then frozen.

Proposition 6 (a frozen cut)**.** *Let the scores of n unlabelled warm-up calls and of a new call be exchangeable with no ties, and let $\hat c$ be the k-th smallest warm-up score. Then $P(S_{n+1} > \hat c) = (n + 1 - k)/(n + 1)$, which for $k = \lceil (1-\beta)(n+1) \rceil$ lies in $\big(\beta - \frac{1}{n+1}, \beta\big]$.*

A cut that follows recent traffic fails exactly when an attack arrives (it catches at most β/ρ of a burst in which a share $\rho > \beta$ violate), while a frozen cut keeps its rate; after a shift of the safe traffic it is set again (Proposition 11, Section C).

Metrics. For geometries: Spearman correlation between two distance matrices over all pairs of contexts (no fitted parameter). For monitors: the AUROC within strata (template, tool or suite), and the share of violations caught when exactly β of the actions are escalated. Intervals are 95% percentile intervals from a bootstrap over contexts, scenarios or episodes. Proofs are in Section B.

Table 1. **Which observable carries the state?** Spearman correlation of each geometry with the model's own hidden geometry, over all pairs of 1,440 held-out contexts; no parameter is fit. Answers: Fisher–Rao distances over 88 questions, exact (log-probabilities) or from 8 answers sampled from the model's own probabilities. Replies: mean embedding of 8 sampled replies. Text: embedding of the input. Best per row in blue; last column: answers − text, 95% interval.

	Answers (ours)				
Model	exact	8 samples	**Replies**	**Text**	**answers − text**
Qwen2.5 7B	**0.74**	**0.74**	0.30	0.28	**+0.46** [0.44, 0.48]
Llama-3.1 8B	**0.74**	0.73	0.32	0.26	**+0.48** [0.46, 0.50]
Mistral 7B	**0.70**	**0.70**	0.22	0.22	**+0.49** [0.47, 0.51]
Qwen3 8B	**0.66**	0.65	0.33	0.30	**+0.36** [0.34, 0.39]
Phi-4 mini	**0.58**	0.57	0.26	0.28	**+0.30** [0.28, 0.33]
OLMo-3 7B	**0.57**	0.55	0.34	0.29	**+0.28** [0.26, 0.30]
Gemma-3 4B	**0.56**	**0.56**	0.44	0.28	**+0.28** [0.26, 0.31]
gpt-oss 20B	**0.41**	0.40	0.25	0.25	**+0.16** [0.14, 0.19]

3. Reconstructing the hidden geometry from answers

We first test the reconstruction where the truth is visible: on open models, whose hidden states we store. **Data.** 4,800 contexts of six kinds (3,360 development, 1,440 confirmation). Two more sets of 1,000 fresh contexts add four new kinds: news, programming, jailbreaks and prompt injections. We ask 88 questions about the message, on eight open chat models from six families. The ground truth is each model's residual stream at the last prompt token, at a layer chosen on development data. All results are on held-out contexts (Section F).

Answers beat the input and the replies. On every model, the answer geometry matches the hidden geometry better than a sentence embedding of the input does (gains +0.16 to +0.49; every lower bound at least 0.141; Table 1). The model's own replies, the other output that an API gives, do little better than the input (0.22–0.44). Agreement rises with the number of questions (from 8 to 88: +0.02 to +0.25), and 8 sampled answers per question keep 97–100% of the exact agreement. On 1,000 fresh contexts of four new kinds, answers beat the input on 6 of 8 models.

The geometry is shared across models. A model's answers match another model's hidden geometry exactly as well as its own. On 1,000 fresh contexts, the mean own-model excess is +0.002 [-0.001, +0.005], inside the equivalence band of ±0.02 that we fixed in advance (Figure 3b); the hidden geometries of different models also agree with each other (0.18–0.80). So what the answers reveal is the geometry that models share, in line with converging representations (Huh et al., 2024; Picozzi, 2026). This finding drives the rest of the paper: the shared part is what any model's answers can read.

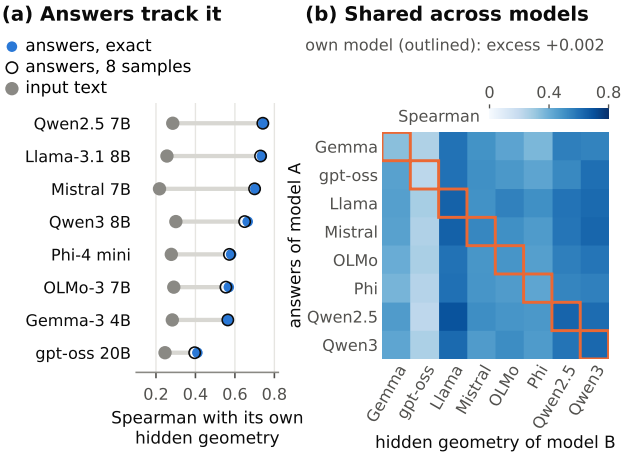


Figure 3. **What yes/no answers reveal of a hidden state** (preregistered). (a) Agreement of each open model's answer geometry with its own hidden geometry, exact and from 8 sampled answers, against the geometry of the input text (Spearman over all pairs of 1,440 held-out contexts; nothing fit). (b) Answers of model A against the hidden geometry of model B on 1,000 fresh contexts: the own-model diagonal (outlined) is not special.

A closed model reveals the same shared geometry. Claude Haiku 5.5 answered 24 questions, 8 times each, on 192 held-out contexts. Two independent halves of its answers give the same geometry (Spearman 0.95 [0.94, 0.96]; bound fixed in advance: 0.7), which carries +0.44 [0.39, 0.49] beyond the input embedding. Its answers agree with open models' hidden geometries at 0.45–0.51.

Its own hidden geometry, by triangulation. Its hidden states are not visible. If, as on open models, no part of its answers is specific to its own state, its own agreement equals its loading on the shared geometry times its hidden states' loading (Proposition 2). Leaving each open model out, as if closed, this estimate predicts its true agreement to within 0.061 [0.051, 0.070] on average (7 of 8 inside the predicted interval). For Claude Haiku 5.5 it gives 0.49, with interval [0.32, 0.53] over the range of open models' loadings.

4. Monitoring, forecasting and preventing violations

Agents and benchmarks. CAPE-Bench (Fayyazi et al., 2026) gives matched tool-use tasks in which one fact decides whether an action is authorised. An environment checker labels each action. We use four open agents, whose decision states we store: Gemma-4 (26B), gpt-oss-20b, Qwen3.8-27B and Muse-Glimmer-30B. We use two closed agents through their public APIs, with reasoning off: Claude Haiku 5.5 (Anthropic) and GPT-6-luna (OpenAI). The second benchmarks are AgentDojo (prompt injection) and τ^2-bench airline.

Table 2. **Monitoring a decision through the agent's own answers** (preregistered; held-out calls). AUROC within strata; best per row in blue; Claude = Claude Haiku 5.5. The read-out of 8 answers is fixed (nothing fit to labels). The trained probe reads the agent's true hidden states (32 principal components, fit on development calls); it is a comparison, not part of the method.

Agent	calls	**8 answers (ours)**	1 answer (self-audit)	judge[d] (reasoning)	trained probe	tuned[c] probe
OPEN AGENTS on CAPE-Bench (within template)						
Gemma-4[e]	600	**0.994**	0.966	0.989	0.980	0.991
gpt-oss-20b	599	0.822	0.758	0.755	0.772	**0.841**
Qwen3.8-27B	373	**0.951**	0.894	0.780	0.826	0.950
Muse-Glimmer-30B	600	0.929	0.935	0.880	0.941	**0.972**
CLOSED AGENTS on CAPE-Bench (no probe can exist)						
Claude, restarts[a]	149	0.909	0.323	**1.000**	–	–
Claude, all[b]	495	0.475	0.306	**0.693**	–	–
GPT-6-luna	560	**0.979**	0.795	–	–	–
AGENTDOJO (within suite) AND τ^2-BENCH (within tool)						
Gemma-4, injection	816	**0.985**	0.975	–	–	–
gpt-oss-20b, injection	821	**0.904**	0.900	–	–	–
Claude, airline writes	360	**0.704**	0.555	–	–	–

[a]Closed: no trained probe exists; a trained reader of the call text scores 1.000 here. [b]Pooled; its mailing-list sends are decided by a fact that it never checked, and they are not flagged. [c]Exploratory: a probe on the true states, with its components and regularisation chosen afterwards by cross-validation on the development calls. [d]Gemma-4 with reasoning on reads the same conversation and gives one holistic violation probability (mean of 4). [e]New task families.

4.1. Monitoring at the call

The read-out. At each proposed call, the agent answers 8 yes/no questions about the call through its own API, 4 times each. The fixed read-out of Section 2 scores the answers. Nothing is fit and no label is used: we fixed the questions and the read-out on development data.

It ranks violations about as well as a probe on the hidden states. On held-out calls, the within-template AUROC is Gemma-4 0.994, gpt-oss 0.822, Qwen3.8 0.951, Muse-Glimmer 0.929 (Table 2). A probe trained on the agent's true decision states, a comparison that needs the weights and labels, scores Gemma-4 0.980, gpt-oss 0.772, Qwen3.8 0.826, Muse-Glimmer 0.941; the mean difference is +0.045 [−0.017, 0.135]. The read-out beats the agent's single self-audit answer by +0.036 [0.006, 0.064]. It does not need the full decision state: the geometry of the 8 answers agrees with that state at only 0.01–0.16, and on open agents about 20% [13, 27] of its ranking power runs through the hidden violation direction. The rest comes from the transcript, which every reader sees.

It beats a reasoning judge on open agents. Gemma-4, with reasoning on, read each conversation and gave one holistic violation probability. It ranked the open agents' violations worse than the read-out (read-out − judge +0.073 [0.021, 0.123]; two-sided test). On Claude Haiku 5.5's actions, which it answered with its reasoning off, the judge is better (Table 2).

With no label, the read-out tells where it works. Its variance beyond its known sampling noise is its SNR (Proposition 4). Over 29 agent–benchmark–stratum cells, the SNR predicts the AUROC (Spearman 0.710 [0.436, 0.870]): cells with SNR > 0 score 0.961 (23 cells), and cells with SNR ≤ 0 score 0.678 (6 cells). The same holds for each question: over eight agent–benchmark pairs, a question's label-free spread beyond its noise predicts how well it ranks violations (mean Spearman 0.580 [0.470, 0.648]). Claude Haiku 5.5's settings have the lowest SNR: its four answers agree on 96–97% of questions.

At the call, another answerer does as well, as the theory implies. The transcript and the call decide the violation (Corollary 1), and the answers read a geometry that models share. So another model that answers the same questions about the same call should match the agent. It does on open agents (own − other +0.012 [−0.018, 0.040]), and it does better on Claude Haiku 5.5 (−0.09 on restarts; −0.29 on τ^2 with reasoning on). Claude Haiku 5.5 itself, reviewing its own calls with reasoning on, reaches 0.918 on τ^2. Monitoring at the call thus works through the agent's own API, but it does not need the agent; forecasting does (Section 4.3).

4.2. Carrying to new benchmarks and closed agents

Prompt injection. On AgentDojo (Debenedetti et al., 2024), the violation is a call that carries out an instruction hidden in a tool output. We wrote 8 questions from that policy before any answer existed. The same read-out ranks the open agents' injected calls within suite, on held-out tasks with no defense, at 0.985 [0.964, 0.998] (Gemma-4, 816 calls, 216 injected) and 0.904 [0.872, 0.932] (gpt-oss-20b, 821 calls, 221 injected); the bar was a lower bound above 0.60.

GPT-6-luna. Through its own API, its answers to the same 8 questions rank its CAPE-Bench violations within template and tool at 0.979 [0.946, 0.998] (560 calls, 66 violations). Its single answer scores 0.795.

Claude Haiku 5.5 on CAPE-Bench. The read-out ranks its unapproved restarts at 0.909 [0.858, 0.952], where its single self-audit answer scores 0.32. Pooled over all its actions, the read-out scores 0.475: a fact that the agent never checked decides its mailing-list sends, so its state holds no warning about them.

Claude Haiku 5.5 on τ^2-bench. On τ^2-bench airline (Barres et al., 2025), Claude Haiku 5.5 breaks the published policy at some of its writes. We wrote 8 questions from the airline policy before any answer existed. The within-tool AUROC is 0.704 [0.536, 0.822] (360 writes, 50 violations, in 7 tasks; bar: lower bound above 0.5), and its single answer to whether the call complies scores 0.555. Per task (exploratory), it ranks the violations of 5 of the 7 tasks (33 of 50) at 0.77–1.00. In the other two, the agent misread the booking, so its state held no warning.

4.3. Forecasting before the call

The pre-call state. Before each step's call, the agent answers 8 questions about its next action; the call does not exist yet.

GPT-6-luna's state forecasts the coming violation. In a first run (exploratory), among calls at the same position, where position cannot help, its pre-call answers ranked the coming violation at 0.927 [0.828, 0.989]. A preregistered replication on 252 fresh episodes, 1047 calls, 68 violations confirms both: within template and tool 0.840 [0.732, 0.946] (position alone 0.777), and among calls at the same position 0.923 [0.811, 0.992] (9 strata holding 27 violations).

Whose forecast is it? Gemma-4 answered the same questions in GPT-6-luna's seat, on the same conversations. Within each kind of task and tool it forecast worse: 0.728, against 0.840 for GPT-6-luna's own answers (+0.111 [0.041, 0.185]). At the same position the difference, +0.08 [−0.02, 0.23], does not separate. *[Pending: fresh episodes of both closed agents: own − other, within template and tool and within position.]*

The forecast lives in the policy, not the self-report. Proposition 3 says the forecast is the agent's own policy, so it can also be read by sampling: draw the agent's next action a few times in side branches and read each drawn call with its own answers. On Claude Haiku 5.5, whose answers about its next action do not rank the coming violation, we test this on fresh episodes: its own draws, read by its own answers, against its self-report and against Gemma-4 sampling in its seat. *[Pending: own draws vs self-report vs another model, within position.]*

Open agents. Among calls at the same step of the same kind of task, the pre-call answers rank the violating calls at 0.67 on Gemma-4 (exploratory) and at 0.669 [0.594, 0.764] pooled over three more agents. On the same steps, 16 answers per question instead of 4 raise the pooled value from 0.65 to 0.73. The deciding fact enters at the call's own step: the answers then rank the call better than one step earlier (+0.16 [0.03, 0.30]). A Kalman forecast from earlier steps, with dynamics identified from unlabelled runs (Proposition 5), beat position on three agents (+0.21 [0.08, 0.32]).

4.4. Preventing violations

Self-policing. Before a non-read call runs, a side branch asks the agent whether the call complies with its policy, and a flagged call goes to review. At one state, a gate prevents the share τ of violating calls that the agent flags. Over many states, it prevents the mean of τ, weighted by where the violations occur (Proposition 12).

Live results. On unseen scenarios (Table 3), Gemma-4 prevented 82% of its violating episodes and reached more goals. Claude Haiku 5.5, asked to check its policy rule by rule, prevented 69% of the violating episodes that its transcript decides, at no cost in goals. Against prompt injection on AgentDojo, the gate cut attack success by 84% and 50%, at no cost in utility.

Table 3. **Self-policing through the agent's own API** (preregistered; held-out scenarios; no other model, no labels). Reduction = 1−gate/plain; better arm in blue.

Agent, set	Violations (%) plain	gate	reduction (95% CI)	Success (%) plain	gate
CAPE-BENCH: episodes with an executed violation; reached the goal					
Gemma-4, all[a]	27.9	**5.0**	**.82** [.76, .88]	33.5	**43.7**
Claude Haiku 5.5, decidable[b]	2.8	**0.9**	**.69** [.51, .88]	40.4	40.1
Claude Haiku 5.5, unchecked[c]	100	92.9	.07 [.00, .17]		
AGENTDOJO: attacks that succeeded; utility					
Gemma-4[d]	11.5	**1.8**	**.84** [.76, .91]	84.5	**86.8**
gpt-oss-20b[d]	24.4	**12.2**	**.50** [.42, .58]	50.5	50.8

[a]666 episodes per arm; a call was held when all four answers said that it does not comply (1.01 reviews per episode; reviewer: CAPE's checker). A second gate, which replaces a flagged call by a fresh draw (up to four), also gave −84%, with success 38.7 and 0.40 reviews per episode. [b]470 scenarios × 2; a call was held when all three answers said that it breaks a rule; 0.31 reviews per episode. [c]21 mailing-list scenarios × 2, decided by a fact the agent never checks; predicted not to fall. [d]Held-out tasks, 2 trials, about 1,300 attacks per arm; spotlighting in both arms; predicted 1.00 and 0.58.

The effect is predictable. Offline flag rates, weighted by where each agent violates, predicted the live reduction of five agents within the registered 0.15 (reductions from 0.01 to 0.63; mean absolute error 0.035; Figure 1d).

A gate on the pre-call state. We ran GPT-6-luna live on fresh seeds, with gates at an equal review rate (1.48 and 1.52 reviews per episode). A gate on the pre-call read-out cut the episodes with a violation from 31.7% to 7.9% (relative reduction 0.750 [0.605, 0.863]); a gate on position cut them to 17.9% (difference +0.312 [0.062, 0.543]). Success did not change (+0.000 [−0.048, 0.048]). The offline share of violating calls that the gate holds, fixed before the run, predicted its effect (0.750; observed 0.750). In all live tests, CAPE's checker stands in for the human reviewer.

Prevention with no reviewer. In the tests above, a held call went to a reviewer. Here nothing judges in the loop: when the agent's own read-out flags a call, the call is held once and the agent is told to re-check its policy and the facts; an identical re-proposal runs. On Claude Haiku 5.5, against random holds at the same rate: *[Pending: violating episodes, none vs self vs random; success.]*

Trained comparisons (Section D). With labels, a probe on an open reader of the transcript is stronger (0.98–0.99 on held-out actions of open and closed agents). But it needs labels for each policy, and it fell to 0.54 on unseen τ^2 tasks. The training-free read-out needs neither, so the two are complementary.

5. The tool

We release a Python package and a website (links in the camera-ready version). Training-free and API-only: `graybox.selfaudit` asks an agent the questions of Section 2 about each proposed call and returns its observable state and the fixed read-out (`observable_state`), or holds a call on its own answer (`SelfAudit`); `graybox-geometry` measures a chat model's observable geometry and states its cost first. **Cost.** With Claude Haiku 5.5 as the answerer, the 32 one-word answers per call (8 questions, 4 answers each) cost $0.0058 per call with prompt caching (360 airline writes for $2.07). An open 26B model on one GPU answered for 360 calls, with reasoning, in 65 minutes. Options that need labels are in Section H.

6. Related work

Information geometry and shared representations. The Fisher metric and its uniqueness are classical (Čencov, 1982; Campbell, 1986; Amari, 2016), as are the arcsine transform (Anscombe, 1948) and classical scaling (Torgerson, 1952); representational similarity analysis compares distance matrices without a fit (Kriegeskorte et al., 2008), and its cross-validated (crossnobis) distance is an unbiased estimator from two independent halves (Walther et al., 2016), the same device as our cross-fit. Representations of different models converge (Huh et al., 2024). Picozzi (2026) show that the Fisher–Rao geometry of next-token distributions is set by behaviour up to symmetries and shared across architectures; Schiekiera et al. (2026) compare behavioural and hidden-state geometry in eight models. Our measurement needs only sampled answers, and we test it against hidden states with a cross-model control.

Black-box access. Logits span a d-dimensional image (Finlayson et al., 2024), and a production model's output projection is recovered up to an orthogonal map (Carlini et al., 2024); Theorem 2 uses the same structure for the gauge-invariant hidden geometry. Answers to questions serve as interpretable embeddings (Benara et al., 2024), detect lies with follow-up questions (Pacchiardi et al., 2024; Cooney et al., 2026; Sam et al., 2025), and forecast behaviour (Levy et al., 2026); open readers probe closed transcripts (Conti et al., 2026), and an open surrogate's log-probabilities audit a black-box agent's tool calls (Zhao et al., 2026); both need a second model, which the grey box does not. We found no prior estimate of how well a closed model's outputs reconstruct its own hidden geometry (Proposition 2).

Questions instead of one verdict. Decomposing an evaluation into yes/no checklist questions makes LLM judges agree better with people than direct scores do (Cook et al., 2024; Lee et al., 2024); we bring decomposition to monitoring an agent's action before it runs, with a fixed read-out and no labels, and test it against probes on hidden states and a holistic reasoning judge.

Probes and monitors. Probes on internals read truth, refusal and deception (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025), and CAPE probes an agent before each action (Fayyazi et al., 2026). Chain-of-thought monitors catch reward hacking that action-only monitors miss (Baker et al., 2025; Korbak et al., 2025; Chen et al., 2025; Arnav et al., 2025); Corollary 1 delimits when the reasoning adds nothing at the Bayes level. An auxiliary open model's hidden states serve as features for a closed model's hallucinations (Sriramanan et al., 2024), and white-box probes are combined with black-box monitors where activations are available (Parrack et al., 2025); we instead read a closed agent through answers to fixed questions, with no training; at the call the answers may be the agent's own or another model's, but before the call the forecast needs the agent's own (Proposition 3).

Self-policing and control. Agents recognize their own misconduct after the fact (Li et al., 2025); deterministic checks of the world before a write recover silent policy violations (Reddy et al., 2026); and resampling suspicious actions limits an adversarial agent (Bhatt et al., 2025). We show that the agent's own audit, through its API and *before* the action, prevents 82% and 69% of the violating episodes that its transcript decides, and offline flag rates predict how much, and confirm the predicted boundary: violations decided by an unchecked fact need such a check of the world. AI control audits a fixed budget of actions (Greenblatt et al., 2024); Propositions 6 and 11 give the budget a label-free, exact and burst-robust form, and the frozen cut's escalation law is the coverage law of split conformal prediction (Vovk et al., 2005; Angelopoulos & Bates, 2023). Self-policing is *untrusted monitoring* (Greenblatt et al., 2024) without an adversary; it cannot stop a model that colludes with itself.

7. Conclusion

A closed agent's API reveals the geometry of its state, not its coordinates, and the geometry it reveals is the one that models share. This one finding sorts what a grey box can do. At a proposed call, the transcript decides the violation, so a fixed read-out of the agent's own answers monitors it with no training, and so can any capable answerer. Before the call, only the agent's own policy carries the forecast; on GPT-6-luna its own answers forecast better than another model's, and a gate on them prevented more than a gate on position, by an amount predicted in advance. Where the theory's label-free noise says a read-out cannot work, or a fact the agent never checked decides the violation, it did not.

Impact Statement

This work monitors closed AI agents before they act, through their own APIs and with no training. The intended effect is safer deployment: in our tests, a gate on the agent's own answers prevented 82% of an open agent's violating episodes and 69% of those that a closed agent's transcript decides. We see four risks. (1) False confidence: the readout cannot see a violation decided by a fact the agent never checked, and its questions cover the kinds of violation they were written for; a new policy needs new questions. (2) Holding calls costs time and, for trained gates, task success (1.8 and 5.8 points). (3) An agent that knows the questions, or an attacker who queries a score, may learn to answer them falsely; the questions and scores should stay hidden from the agent, and self-policing is no defense against a model that colludes with itself. (4) Reading a model's state from outside could help an attacker study a closed model; the method uses only ordinary answers and recovers geometry, not weights or private data. The code is released under the MIT license, and the browser tool does not upload the user's logs.

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A. Limitations

- **Benchmarks.** Most agent results use CAPE-Bench, whose tasks are synthetic and matched so that one fact decides each action. AgentDojo and τ^2-bench airline test the read-out on other policies, but on fewer agents.
- **Questions follow the policy.** The 8 questions were written from CAPE-Bench's kinds of violation. Another policy needs its own questions, written before any answer exists, as we did for AgentDojo and τ^2-bench.
- **A fact that the agent never checked.** If a violation depends on a fact that is not in the transcript and that the agent never looked up, the agent's state holds no warning, and the read-out cannot flag it (Claude Haiku 5.5's mailing-list sends; two of seven τ^2 tasks, where the agent misread the booking). Such violations need a check of the world.
- **Nearly deterministic answers.** When an agent gives the same answer to almost every question every time, the read-out has little variance beyond its noise. The label-free SNR (Proposition 4) flags these settings before any label exists; Claude Haiku 5.5's settings have the lowest SNR.
- **A forecast cannot beat the agent's own randomness.** Where the agent samples its action at the call (the same state leads to a violation in one run and not in another), no reading before the call can forecast more than the rate at that state (Proposition 3).
- **A closed model's own geometry is estimated, not measured.** The triangulation assumes that the closed model, like the open ones, has no own-model excess, and that its hidden states load on the shared geometry like an open model's.
- **The reviewer.** In the live tests, CAPE's checker stands in for the human who reviews a held call. A human reviewer is slower and makes mistakes, so live prevention with people can be lower.
- **Adversaries.** An agent that knows the questions, or an attacker who can query the score, may learn to answer falsely. Self-policing is untrusted monitoring without an adversary; it is no defense against a model that colludes with itself. The questions and scores should stay hidden from the agent.
- **Cost.** The read-out asks 32 one-word questions per call (8 questions, 4 answers each). With prompt caching this cost $0.0058 per call on Claude Haiku 5.5, and it adds one round of parallel calls of latency.
- **Many tests.** We ran many preregistered tests and apply no correction for multiple tests. Each claim rests on the primary endpoint of its own test (Section L).

B. Proofs

Exact recovery from log-probabilities

The model-specific final geometry from log-probabilities. Theorem 2 recovers a model's own final geometry exactly from full log-probabilities. On Qwen3-0.6B ($d = 1{,}024$), from 1.15 million self-generated contexts, the recovered cosines of 2,000 held-out contexts correlate with the true ones at 0.30 and 0.36 (at $0.25\times$ and $0.5\times$) below $d(d+1)/2$ contexts (at $0.75\times$ and $0.9\times$ the minimum-norm solution is indefinite and the recovery fails); at $1.1\times$ the correlation is 0.9915, and at $2\times$ the median error is 7×10^{-6} (maximum 1.5×10^{-4}) (Figure 4b; preregistered). Closed APIs do not return full log-probabilities.

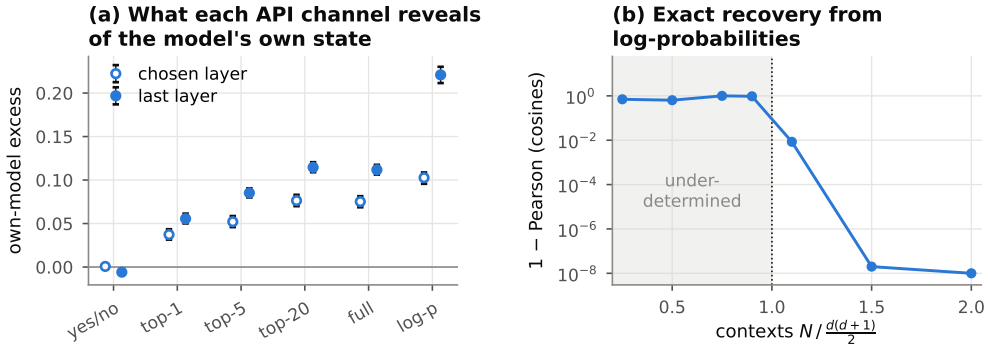


Figure 4. (a) Own-model excess for each API channel, against the last layer and the chosen layer, on 1,000 fresh contexts (preregistered; 95% intervals). (b) Exact recovery of the final hidden geometry from full log-probabilities (Qwen3-0.6B, 2,000 held-out contexts) against the number of contexts; the threshold is $d(d+1)/2$ (preregistered).

Where an API returns full log-probabilities, the final state is not estimated but recovered, by the construction of Carlini et al.

(2024, App. H).

Theorem 2 (exact recovery; after Carlini et al.)**.** *Let the logits be Wu, with $u = x/\mathrm{rms}(x)$ the normalised final state, the normalisation gain folded into W, and no output bias. The centred log-probabilities $c(x)$ lie in a d-dimensional subspace; write $b = Q_d^\top c$ for an orthonormal basis Q_d of it. Then $u = Kb$ for an unknown invertible K, and $\|u\|^2 = d$ gives $b_i^\top S b_i = d$ with $S = K^\top K$: a linear system in S. If the $b_i b_i^\top$ span the symmetric matrices (generically, $N \geq d(d+1)/2$ contexts), S is unique, and the Gram matrix of the final states, $b_i^\top S b_j = u_i^\top u_j$, is exact for every new context. The states follow up to an orthogonal map: the gauge group.*

The subspace (Finlayson et al., 2024) and the ellipsoid fit that removes all but an orthogonal ambiguity (Carlini et al., 2024, App. H) are known; we state the consequence for the gauge-invariant geometry and test its sharp transition at $d(d+1)/2$ contexts. Recovery holds up to the normalisation's ϵ, and near the threshold the system is badly conditioned.

B.1. The observable geometry

Proof of Theorem 1. (a) The Fisher information of $\mathrm{Bernoulli}(p)$ is $1/(p(1-p))$ and $\mathrm{d}\vartheta/\mathrm{d}p = 1/\sqrt{p(1-p)}$, so the metric is $\mathrm{d}\vartheta^2$. The Fisher metric of a product of independent factors is the sum of the factors' metrics, so in ϑ the family is the box $[0,\pi]^m$ with the Euclidean metric. The box is convex, so geodesics are segments and the distance is Euclidean. Check for one factor: the Bhattacharyya coefficient of $\mathrm{Bernoulli}(\sin^2\alpha)$ and $\mathrm{Bernoulli}(\sin^2\beta)$ is $\cos(\alpha-\beta)$, and $2\arccos\cos(\alpha-\beta) = |\vartheta - \vartheta'|$. (b) Invariance: d_Q is a function of $(P_x, P_{x'})$ only, and Proposition 1 leaves every output law unchanged. Uniqueness of the Fisher metric on the simplex over the finite set $\{0,1\}^m$, under Markov morphisms, is Čencov's theorem. d_Q is the intrinsic geodesic distance of the submanifold of independent laws; the geodesic distance of the whole simplex between two product laws is $2\arccos$ of their Bhattacharyya coefficient, $2\arccos\prod_j \cos\frac{\vartheta_j - \vartheta'_j}{2}$, which is a different function of the same angles. (c) $\mathrm{d}\vartheta_j = \Psi'_j/\sqrt{\Psi_j(1-\Psi_j)}$ and $\Psi'_j = \Psi_j(1-\Psi_j)\nabla a_j^\top \Delta$, so $\mathrm{d}\vartheta_j = \sqrt{\Psi_j(1-\Psi_j)}\,\nabla a_j^\top \Delta$; summing the squares gives $\Delta^\top G_Q \Delta$, and the remainder is third order by Taylor's theorem. (d) $\hat\vartheta$ is increasing in k and k is stochastically increasing in p, so μ_n is strictly increasing; $\mu_n \to \vartheta$ by the law of large numbers and continuity. The halves are independent, so $\mathbb{E}[(a_A - a'_A)(a_B - a'_B)] = (\mathbb{E}a_A - \mathbb{E}a'_A)(\mathbb{E}a_B - \mathbb{E}a'_B) = (\mu_n(\Psi) - \mu_n(\Psi'))^2$ for each question; sum over questions. This needs the same n in every cell: with $n_A(x) \neq n_A(x')$ the factor is $\mu_{n_A(x)}(\Psi) - \mu_{n_A(x')}(\Psi')$, not a difference of one function. Clipping a negative $\hat D^2$ at 0, as needed for distances, adds a non-negative bias. Classical scaling recovers a Euclidean configuration from its distance matrix up to a rigid motion (Young–Householder; Torgerson). Note that the familiar large-n variance $1/(n+\frac{1}{2})$ of the Anscombe angle fails near $p \in \{0,1\}$ and its bias is $\cot\vartheta/(4n)$, so subtracting a variance term is not exact; the cross-fit needs neither. □

Proof of Theorem 2. With $P = I - \mathbf{1}\mathbf{1}^\top/V$, the log-softmax removes a constant per context, so $c = PWu$ and every c lies in the column space of PW, of dimension d when PW has full column rank. Let Q_d be an orthonormal basis of it and $b = Q_d^\top c = Mu$ with $M = Q_d^\top PW$ invertible; then $u = Kb$ with $K = M^{-1}$. $\|u\|^2 = b^\top K^\top K b = d\,\overline{x^2}/(\overline{x^2} + \epsilon)$, which is d up to the normalisation's ϵ. The map $S \mapsto (b_i^\top S b_i)_i$ is linear on the $d(d+1)/2$-dimensional space of symmetric matrices; it is injective iff the $b_i b_i^\top$ span that space, which for b_i in general position needs and generically holds with $N \geq d(d+1)/2$. Then $S = K^\top K$ is the unique solution and $B^\top S B = U^\top U$. Any L with $L^\top L = S$ satisfies $L = OK$ for an orthogonal O, so the states are fixed up to O, the group of Proposition 1. □

The variance-subtraction rule is not exact. The rule estimates the squared distance by $\|\hat\vartheta(x) - \hat\vartheta(x')\|^2 - 2m/(n+\frac{1}{2})$. Exact binomial sums show that the bias of $\hat\vartheta$ is $\cot\vartheta/(4n) + O(n^{-2})$ (at $n = 32$: 0.0102 vs 0.0104 at $p = 0.1$), that the variance is below $1/(n+\frac{1}{2})$ near $p \in \{0,1\}$ ($0.68/(n+\frac{1}{2})$ at $p = 0.1$, $n = 8$), and that with n samples the angles cannot leave $[2\arcsin\sqrt{(3/8)/(n+3/4)},\ \pi - 2\arcsin\sqrt{(3/8)/(n+3/4)}]$, so large distances between saturated answers contract. The cross-fit of Theorem 1(d) replaces the rule. The preregistered endpoints use the subtraction rule, as specified. On Claude Haiku 5.5 the cross-fit lowers the negative eigenvalue mass of classical scaling from 15% to 2%; on open models it keeps the same rank agreement with the hidden geometry (exploratory).

The solver. We whiten b on the training contexts (any invertible map of b is allowed) and solve the least-squares problem for symmetric S by conjugate gradient on the normal equations, from $S = 0$, which gives the minimum-norm solution when $N < d(d+1)/2$. On synthetic heads with known K ($d = 16, 48$) the cosine error is 10^{-10} to 10^{-12} above the threshold and 0.04–0.09 below it.

Remarks on Theorem 1. On the whole simplex over $\{0,1\}^m$ the Fisher–Rao distance between two independent answer laws is $2\arccos\prod_j \cos\frac{\vartheta_j-\vartheta'_j}{2}$; on the submanifold of independent laws it is d_Q of part (a). In part (c), s is the whole cache that the questions read, so its prediction for one layer's vector is heuristic. In part (d), the familiar rule of subtracting the large-n variance $1/(n+\frac{1}{2})$ is not exact: near $\Psi \in \{0,1\}$ the variance is smaller and the bias is $\cot\vartheta/(4n)$; an answer that is neither yes nor no, or clipping negative estimates at 0, breaks the exactness. The answer law is conditional on a yes or no answer.

Notation is that of Section 2. $\sigma(X)$ is the σ-field generated by X. A score is *strictly proper* if its expectation is uniquely maximised by the true conditional probability.

Proposition 7 (behavioural sufficiency)**.** *Let $Y = g(\tau)$ be defined by the future τ that the agent and the environment generate from context x, and let $\phi(x)$ carry no information about the future sampling noise given x. Then $\pi^*(x) = P(Y=1 \mid x)$ has a maximal ROC curve and a minimal risk under every strictly proper score among all forecasts $f(\phi(x))$.*

Proof. By the assumption and the tower property, $\mathbb{E}[Y \mid \phi(X)] = \mathbb{E}[\pi^*(X) \mid \phi(X)]$. Under a strictly proper score the Bayes act given $\sigma(X)$ is $\pi^*(X)$, and $\sigma(\phi(X)) \subseteq \sigma(X)$; less information cannot raise the expected score. For the ROC curve, by the Neyman–Pearson lemma the most powerful test of each size thresholds the likelihood ratio $p(x \mid Y{=}1)/p(x \mid Y{=}0)$, a strictly increasing function of $\pi^*(x)$; every test based on $f(\phi(X))$ is a test based on X. The same holds within any stratum that is a function of X, hence for the within-stratum AUROC. □

Proposition 1 (gauge). Write a pre-norm residual network as $h^{(\ell+1)} = h^{(\ell)} + W_{\text{out}}^{(\ell)} F^{(\ell)}\big(W_{\text{in}}^{(\ell)}\, N(h^{(\ell)})\big) + b_{\text{out}}^{(\ell)}$, with embedding $h^{(0)} = E(x)$, output logits $U\,N(h^{(L)})$ and $N(h) = h/\mathrm{rms}(h)$ after folding the normalisation gains into W_{in} and U. For orthogonal G, $\mathrm{rms}(Gh) = \mathrm{rms}(h)$, so $N(Gh) = G\,N(h)$. Replace $E \mapsto GE$, $W_{\text{out}} \mapsto GW_{\text{out}}$, $b_{\text{out}} \mapsto Gb_{\text{out}}$, $W_{\text{in}} \mapsto W_{\text{in}}G^\top$ and $U \mapsto UG^\top$. By induction on ℓ the new residual stream is $Gh^{(\ell)}$, since $W_{\text{in}}G^\top N(Gh) = W_{\text{in}}N(h)$ and the new update is G times the old one; the logits $UG^\top N(Gh^{(L)}) = UN(h^{(L)})$ are unchanged, so the output distribution is unchanged on every input. At a layer read only through linear maps (no normalisation in between), the same argument holds for every invertible G with G^{-1} in place of $G^\top$. Any estimator of hidden coordinates that uses only outputs takes the same value on both networks, so it cannot distinguish h from Gh; a probe $w^\top h$ on one network equals $(G^{-\top}w)^\top(Gh)$ on the other. *Other normalisations.* LayerNorm subtracts the mean, $N(h) = Ph/\mathrm{rms}(Ph)$ with $P = I - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$; $N(Gh) = GN(h)$ holds for orthogonal G with $G\mathbf{1} = \mathbf{1}$, which still form a group of dimension $(d-1)(d-2)/2$. A normalisation *after* a block, $\mathrm{diag}(\gamma)N(\cdot)$ added to the stream, needs $G\,\mathrm{diag}(\gamma) = \mathrm{diag}(\gamma)\,G$: G must commute with the gains, so for distinct gains only sign flips remain, and for equal gains any orthogonal map within the block of equal gains. From exact full-vocabulary log probabilities the final hidden state is identified only up to such a G (Carlini et al., 2024). □

Fisher–Rao coordinates are proved with Theorem 1 above.

Proposition 13 (battery design). Each chosen question splits at least one class, so the number of classes grows by at least one per question and starts at one; the loop stops when no question in the pool splits a class. At that point two contexts in the same class agree on every pool question (otherwise that question would split their class), and two contexts in different classes disagree on a chosen question; so the final partition is the one that the whole pool induces, reached with at most (number of its classes $-\,1$) questions. If the answers come from a finite automaton with minimal automaton M, two prefixes give the same answer to every suffix iff they reach the same state of M (Myhill–Nerode); when the pool holds a distinguishing suffix for every pair of states, the pool partition is the partition by minimal state. States that never change an output are behaviourally inert, so no black-box method identifies more. On a program compiled into a transformer (a bracket-balance automaton: 21 reachable states plus a dead state), the algorithm chose 21 questions from a pool of 2,101 and separates all 22 classes. □

Proposition 8 (access and replica). (a) $\phi_t = \Phi_\theta(Z_t)$ is $\sigma(Z_t)$-measurable, so $\sigma(Z_t, \phi_t) = \sigma(Z_t)$ (for every label), and Proposition 7 applied with $X = Z_t$ gives the first claim. Under log loss the Bayes risk given a σ-field $\mathcal{G}$ is the conditional entropy $H(Y \mid \mathcal{G})$, and $H(Y \mid V_t) - H(Y \mid V_t, H_t) = I(Y; H_t \mid V_t)$. (b) By the law of total covariance, $\mathrm{Cov}(\phi) = \mathbb{E}\,\mathrm{Cov}(\phi \mid \psi) + \mathrm{Cov}\big(\mathbb{E}[\phi \mid \psi]\big)$, and the first term is positive semidefinite. For component k, $\mathrm{Var}(\phi_k) = \mathbb{E}(\phi_k - \hat\phi_k)^2 + \mathrm{Var}(\hat\phi_k)$, so $\mathrm{Var}(\hat\phi_k) = R_k^2\,\mathrm{Var}(\phi_k)$. For the best linear predictor $\hat\phi_k = a_k^\top\psi + b_k$ the residual is uncorrelated with $\hat\phi_k$ (the normal equations), and the same decomposition holds in sample. On fresh steps neither identity is exact: Figure 7d shows the ratio closely tracking R_k^2 on 128 components (correlation 0.99, slope 0.89), above R_k^2 on 80 of them. (c) Conditional expectation is linear: $\mathbb{E}[w^\top\phi \mid \psi] = w^\top\mathbb{E}[\phi \mid \psi]$. □

Proposition 9 (the replica Rao–Blackwellises a probe). $s = w^\top\phi$ is a function of ϕ, so $Y \perp\!\!\!\perp \phi \mid \psi$ gives $Y \perp\!\!\!\perp s \mid \psi$

and hence $\mathbb{E}[s \mid \psi, Y] = \mathbb{E}[s \mid \psi] = \hat{s}$. For ℓ convex in its first argument, conditional Jensen gives $\mathbb{E}[\ell(s, Y) \mid \psi, Y] \geq \ell(\mathbb{E}[s \mid \psi, Y], Y) = \ell(\hat{s}, Y)$; take expectations. For a pair of independent draws $(\phi_i, \psi_i, Y_i = 1)$ and $(\phi_j, \psi_j, Y_j = 0)$, $\mathbb{E}[s_i - s_j \mid \psi_i, \psi_j, Y_i, Y_j] = \hat{s}_i - \hat{s}_j$ by the same conditional independence and the independence of the draws, and Jensen applies to $\ell(s_i - s_j)$. The AUROC is the expectation of the non-convex step $\mathbf{1}\{s_i > s_j\}$, so the result covers its convex surrogates (logistic, exponential, hinge), not the AUROC itself. □

Corollary 1 (sufficiency of the proposed action). Let $A_t = a(Z_t^-, \xi)$ with $\xi \perp\!\!\!\perp (Z_t^-, W)$, and let V_t be a function of Z_t^-.

1. $\xi \perp\!\!\!\perp (Z_t^-, W)$ gives $\xi \perp\!\!\!\perp W \mid Z_t^-$, hence $\xi \perp\!\!\!\perp W \mid (Z_t^-, V_t)$; with $W \perp\!\!\!\perp Z_t^- \mid V_t$, contraction gives $W \perp\!\!\!\perp (Z_t^-, \xi) \mid V_t$.
2. A_t is a function of (Z_t^-, ξ), so $W \perp\!\!\!\perp (Z_t^-, A_t) \mid V_t$, and by weak union $W \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$.
3. $Y = g(A_t, V_t, W)$, so $Y \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$: $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$.
4. A monitor that reads internals at or before the action, and possibly the action, sees a σ-field inside $\sigma(Z_t^-, A_t)$. Under a strictly proper score the Bayes risk given a coarser σ-field is no smaller, and by step 3 the Bayes act given $\sigma(Z_t^-, A_t)$ is $P(Y \mid V_t, A_t)$; so its Bayes risk is at least that of the best monitor of (V_t, A_t). □

The inequality can be strict: if A_t is random given Z_t^- and Y depends on which action is sampled, a probe read before the action must average over the actions. *Canonical form.* If moreover $Y = g(\kappa(A_t), V_t, W)$ for a function κ of the action, the same steps with $\kappa(A_t)$ in place of A_t show that $(V_t, \kappa(A_t))$ is sufficient, and every monitor of $(V_t, \kappa(A_t))$ is invariant to any change of A_t that keeps $\kappa(A_t)$. The premise is about the label. In our attacked family it held: the checker labels a send by its recipient.

Proposition 10 (a ceiling under rewording). Write an action as (κ, t), with t its free text, and let $T_0(V, \kappa)$ be the set of texts that safe actions with context V and decisive part κ use. Define $m(V, \kappa) = \inf_{t \in T_0(V,\kappa)} s(V, \kappa, t)$, and $m = +\infty$ where T_0 is empty. On a safe action $t \in T_0(V, \kappa)$, so $m \leq s$ and $m \geq c$ implies $s \geq c$: m escalates no more safe actions than s. Under the attack, a violation with (V, κ) is rewritten to the text in $T_0(V, \kappa)$ with the lowest score (allowed by the assumption; take a sequence if the infimum is not attained), so its attacked score is $m(V, \kappa)$, and it is escalated iff $m(V, \kappa) \geq c$. As m depends only on (V, κ), the attack does not change it, and its share of violations caught on clean traffic equals the share that s catches under the attack. Where T_0 is empty the attacker cannot imitate a safe action, and $m = +\infty$ escalates. □

Proposition 6 (a frozen cut). For exchangeable scores without ties, the rank of S_{n+1} among $S_1, \ldots, S_{n+1}$ is uniform on $\{1, \ldots, n+1\}$. $S_{n+1} > S_{(k)}$ holds iff its rank exceeds k, which has probability $(n+1-k)/(n+1)$; with $k = \lceil (1-\beta)(n+1) \rceil$ this lies in $(\beta - \frac{1}{n+1}, \beta]$. For i.i.d. scores with a continuous distribution function F, a new score exceeds $\hat{c}$ with probability $1 - F(\hat{c})$ given the warm-up; $F(S_{(k)})$ is the k-th order statistic of n uniforms, $\mathrm{Beta}(k, n+1-k)$, so over warm-up draws $1 - F(\hat{c}) \sim \mathrm{Beta}(n+1-k, k)$, with mean $(n+1-k)/(n+1)$ and variance $k(n+1-k)/\big((n+1)^2(n+2)\big)$; for $n = 500$, $\beta = 0.05$: $k = 476$, mean 0.0499, sd 0.0097. The shipped cut is the linearly interpolated empirical quantile at $1 - \beta$, which lies in $[S_{(m)}, S_{(m+1)}]$ with $m = \lfloor 1 + (n-1)(1-\beta) \rfloor$. Its escalation probability is therefore in $\left[\frac{n-m}{n+1}, \frac{n+1-m}{n+1}\right]$, and $(n-1)(1-\beta) \leq m \leq 1 + (n-1)(1-\beta)$ puts it in $\left[\beta - \frac{2\beta}{n+1}, \beta + \frac{2(1-\beta)}{n+1}\right]$, within $\frac{2}{n+1}$ of β (for $n = 500$: between 0.0499 and 0.0519). □

Proposition 11 (shift). For every c, $Q(S > c) - P(S > c) = F_P(c) - F_Q(c)$, whose absolute value is at most $K(P, Q)$. Apply it at $c = \hat{c}$ and combine with Proposition 6; apply it to the laws of the violations' scores for the share caught. □

Proposition 11 (bursts). A cut at the top β of a window of N actions escalates $\lceil \beta N \rceil$ of them; the window holds ρN violations, so at most $\beta N/(\rho N) = \beta/\rho$ of them are escalated (up to rounding). A frozen cut $\hat{c}$ escalates a violation with probability $P(S \geq \hat{c} \mid Y{=}1)$, the monitor's true-positive rate at $\hat{c}$, which does not depend on how many violations the traffic holds when the score laws do not depend on ρ. A rolling window that holds only part of a burst has a smaller effective share than the burst, so it can catch more than β/ρ of the burst's violations; the bound is on the window. □

C. Theory for monitoring agents

This appendix holds the theory that the agent results use. The main text states Corollary 1, a short form of Proposition 6, and the shift and burst bounds.

The reader replicates the agent's state. A *reader* is an open model R that reads the same visible tokens and the proposed action; $\psi_t = R(V_t, A_t)$ is its hidden state at a fixed layer, and ϕ_t is the agent's state at the decision token.

Proposition 8 (access and replica). *(a) $\sigma(Z_t, \phi_t) = \sigma(Z_t)$. For every label Y, the best monitor that reads (Z_t, ϕ_t) is no better than the best that reads Z_t, under every strictly proper score and at every point of the ROC curve. Under log loss, the*

best monitor that reads Z_t beats the best that reads V_t by exactly $I(Y; H_t \mid V_t)$. (b) The L^2-optimal replica $\hat{\phi} = \mathbb{E}[\phi \mid \psi]$ satisfies $\mathrm{Cov}(\hat{\phi}) \preceq \mathrm{Cov}(\phi)$ and, for every component k, $\mathrm{Var}(\hat{\phi}_k) = R_k^2 \, \mathrm{Var}(\phi_k)$, with $R_k^2 = 1 - \mathbb{E}(\phi_k - \hat{\phi}_k)^2 / \mathrm{Var}(\phi_k)$. The same identity holds for the best linear replica, in sample. (c) For every linear readout w, $\mathbb{E}[w^\top \phi \mid \psi] = w^\top \hat{\phi}$: a probe applied to the replica is the conditional expectation of the same probe on the true state.

Part (a) says why a replica is possible: with reasoning off, ϕ_t is a function of tokens that an outside model can also read. Part (b) predicts the shape of the replica, and part (c) says what a white-box readout computes on it.

Proposition 9 (the replica Rao–Blackwellises a probe)**.** *Let $s = w^\top \phi$ be a probe on the agent's state and $\hat{s} = \mathbb{E}[s \mid \psi]$ its readout on the L^2-optimal replica. If $Y \perp\!\!\!\perp \phi \mid \psi$, then $\mathbb{E}\,\ell(\hat{s}, Y) \le \mathbb{E}\,\ell(s, Y)$ for every loss $\ell(s, y)$ that is convex in s. The same holds for every convex pairwise loss $\mathbb{E}\,\ell(s_i - s_j)$ of independent violating and safe actions i, j, the convex surrogates of the AUROC.*

We estimate $\hat{\phi} = M\psi + b$ by ridge regression on paired steps of an open agent, on the top principal components of its state; the proposition applies to this estimate only as far as the linear map approximates the conditional mean.

Robustness to rewording. Let $\kappa(A)$ be the decisive part of an action (its tool and targets) and let the free text be the rest.

Proposition 10 (a ceiling under rewording)**.** *Suppose that for every violating action an attacker can reach a still-violating action with the same $\kappa(A)$ and any free text that safe actions with the same $(V, \kappa(A))$ use, and that the attacker picks, among these, the text with the lowest score of the monitor. Then for every monitor s and cut c there is a monitor m of $(V, \kappa(A))$ alone that escalates no more safe actions than s and, on clean traffic, catches at least the share of violations that s catches under the attack.*

Under rewording, no monitor's operating point beats the clean ROC curve of the best monitor of the decisive fields.

Self-policing: a prevention law. The agent is asked in a side branch whether its proposed call complies; a flagged call is replaced by a fresh draw, at most M draws, and after M flags goes to review ($M = 1$ is the gate).

Proposition 11 (shift and bursts)**.** *If deployment scores follow Q and warm-up scores follow P, a frozen cut escalates within $K(P, Q) + \frac{2}{n+1}$ of β, with K the Kolmogorov distance. A cut set at the top β of a window in which a share $\rho > \beta$ of the actions are violations catches at most β/ρ of them; a frozen cut catches each violation with probability $P(S \ge \hat{c} \mid Y = 1)$, whatever ρ is.*

The full statements of the monitor's algorithm and a ceiling on robustness to rewording (Proposition 10) follow.

Proposition 12 (prevention law)**.** *At a state let R be the agent's probability of proposing a violating call, and let the audit flag a violating call with probability τ and a safe one with probability φ, independently across draws. With $f = R\tau + (1 - R)\varphi$, a violation runs without review with probability $R(1 - \tau)\,(1 - f^M)/(1 - f)$, and the step goes to review with probability f^M.*

Proof. Draw j is the first unflagged draw with probability $f^{j-1}(1 - f)$; given that it is unflagged, it violates with probability $R(1 - \tau)/(1 - f)$. Summing over $j \le M$ gives the first claim; all M draws are flagged with probability f^M. □

Every term is measurable through the agent's API: R by resampling it, τ and φ by auditing it. For a gate whose reviewer blocks flagged violations, the share of violations prevented at one state is τ when a blocked violation is not retried. Over states of kinds k with violation shares w_k and flag rates τ_k, it is $\sum_k w_k \tau_k$. Gemma-4's development audit flagged $\tau = 0.883$ of its violating calls when all four answers said no (the gate's cut; 0.966 when a majority did); the live gate prevented 0.82 [0.76, 0.88] of its violating episodes.

The frozen cut, in full. Proposition 6 also gives the long-run escalation rate: for i.i.d. scores with continuous distribution function F, $1 - F(\hat{c})$ follows $\mathrm{Beta}(n + 1 - k, k)$ over warm-up draws, and the interpolated empirical $(1 - \beta)$-quantile has escalation probability within $\frac{2}{n+1}$ of β. With $n = 500$ and $\beta = 0.05$ the escalation rate has mean 0.050 and standard deviation 0.0097.

Battery design. In our earlier tests of the interrogation state (an Othello-playing transformer; preregistered, Section L), the battery was designed from black-box answers alone: from a pool of candidate questions, add the one that splits the most classes of contexts with equal answers, and stop when no candidate splits a class.

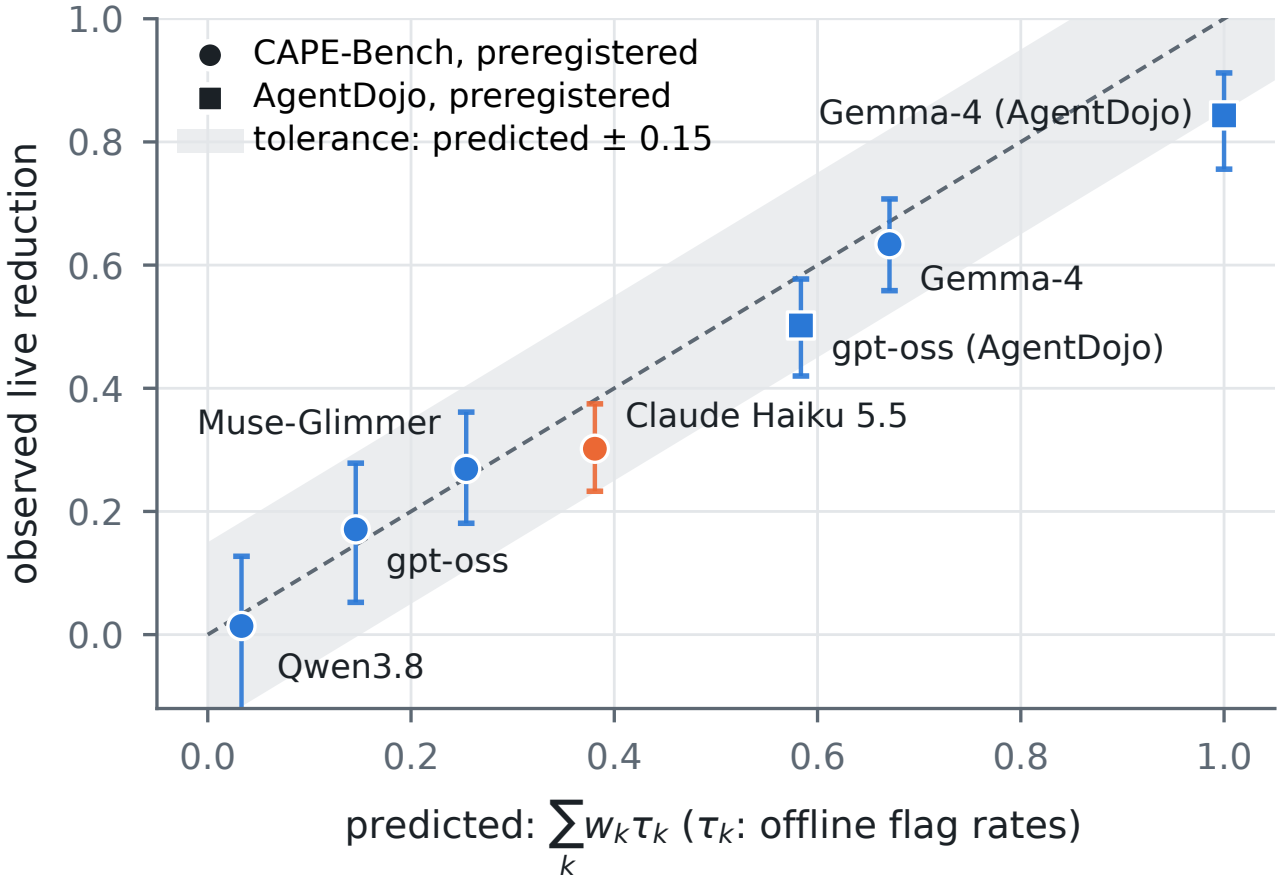


Figure 5. **The prevention law predicts live self-policing.** Live reduction in violating episodes (95% interval) vs the prediction from offline flag rates per kind of task, weighted by where the agent violates. Dots: the composition-weighted law (preregistered; all five within ± 0.15). Squares: prompt injection on AgentDojo (preregistered).

Proposition 13 (battery design)**.** *This greedy rule stops with the partition of the contexts that the whole pool induces and asks at most (number of its classes -1) questions. If the model's answers come from a finite automaton and the pool holds a distinguishing suffix for every pair of its states, this partition is the minimal (Myhill–Nerode) state, and no black-box method can identify more.*

D. Trained comparisons: the reader monitor in full

Trained comparisons. This appendix reports monitors that are *trained*, as comparisons for the training-free method. With labels, an open reader of the transcript is a monitor: a logistic probe on its state, trained only on 17,623 Gemma-4 steps, scores every agent's actions (Table 4); baselines are the agent's own internals (where open), TF-IDF with the same labels, a zero-shot 70B judge, a reasoning judge given the policy, and the action's tool alone. **The transcript is enough.** Corollary 1 says that the agent's internals cannot beat a monitor of the transcript and the action. A probe on Gemma-4's own internals scores 0.841 at the start of the turn and 0.979 after its reasoning; the reader probe, trained on the same labels, scores 0.994 ($+0.015$ [0.007, 0.026] over the internals) and is non-inferior on unseen task families (Table 4). Given the action, adding the agent's hidden reasoning changes Gemma-4's AUROC by -0.0014 [-0.0046, $+0.0014$]. On three other reasoning agents (preregistered), the reasoning lowered transfer from Gemma-4 by -0.033 [-0.046, -0.017]. Fit on the agent's own steps, the reader's state replicates 46–66% of the variance of the agent's top 128 decision-state components on four architectures (Sections C and J). Over word counts it adds robustness to shift: TF-IDF ties it in distribution but is at chance on AgentDojo (0.503 vs 0.781).

Two closed agents. Nothing was fit on the closed agents. At a cut set at the top 5% of the agent's own unlabelled traffic,

Algorithm 1 The grey-box monitor

Require: open-agent steps $\{(V_i, A_i, Y_i)\}$; reader R, layer ℓ; budget β; warm-up length n
$x_i \leftarrow \text{render}(V_i, A_i)$: instructions, task, earlier calls and results, proposed action
$\psi_i \leftarrow R_\ell(x_i)$ at the last token {the reader's state}
fit f: standardise $\rightarrow$ PCA(256) $\rightarrow$ logistic (C=0.05) on (ψ_i, Y_i) {never on the test agent}
for the first n actions A_t of the monitored agent **do**
  record $s_t \leftarrow f(R_\ell(\text{render}(V_t, A_t)))$; run A_t {no labels}
end for
$\hat{c} \leftarrow$ the $(1-\beta)$-quantile of $s_{1..n}$; freeze it {Propositions 6 and 11}
for each later proposed action A_t **do**
  if $f(R_\ell(\text{render}(V_t, A_t))) \geq \hat{c}$ **then** block A_t, or hold it for review **else** run A_t
end for

the monitor catches 94.1% [85.8, 97.7] of Claude Haiku 5.5's 68 violations (zero-shot judge: 0%; TF-IDF: 63.8%) and 98.5% [91.9, 99.7] of GPT-6-luna's (judge 37.9%; TF-IDF 49.0%; Wilson intervals). Claude Haiku 5.5's violations are two behaviours: 42 test sends to a mailing list with thousands of external members it never checked, and 26 production restarts without approval. A prior that knows only each template's and tool's violation rate, fit out of fold, catches 91.6% [82.1, 95.9] (AUROC 0.992; reader 0.989), so on Claude Haiku 5.5 the result shows that the monitor flags these two behaviours, not that it reads the deciding facts. GPT-6-luna's 66 violations span six templates, each with safe calls, and there the monitor beats that prior (98.5% vs 86.4% [76.1, 92.7]), as on Gemma-4. Score scales do not transfer between agents (a cut at 0.5 catches 3% on Claude Haiku 5.5), rankings do: the cut must come from the agent's own traffic. Frozen after 500 actions, it keeps catching 94% under bursts in which half the actions violate; a cut that moves over 5,000 actions falls to 36% (Proposition 11, Figure 8a).

Live prevention. We ran the monitor as a server in each closed agent's tool loop, on the same scenarios with and without the gate. In all four preregistered live tests the gate removed nearly all executed violations (Table 5 and Figure 6), and no agent executed a violation after a block. Holding an escalated action for a reviewer kept the safety gain. The round trip was 89 ms on GPT-6-luna and 80 ms on Claude Haiku 5.5 (median; 95th percentiles 215 and 139 ms).

An agent that knows it is monitored, and an attacker who queries the score. Told that a monitor reviews every call, Claude Haiku 5.5 violated as often; the monitor caught 97.1% at the frozen cut, which then escalated 15.2% of actions (88.6% at a 5% budget; preregistered); a cut set again on its unlabelled traffic removed every executed violation live (Table 5). An attacker who queries the score and rewrites a violating call's free text lowered the catch from 96% to 29% (Section A): the score must stay hidden from the agent.

Labels of a new policy transfer from open agents. Prompt injection on AgentDojo is absent from our training data; its labels from two open agents transfer to the closed GPT-6-luna (AUROC 0.922 vs the judge's 0.787; Table 4).

E. A trained comparison: recreating the decision state

This comparison is not part of the method: it trains linear maps and probes on open agents. Let O_t be what the API yields at the decision: an open reader's state ψ_t of the same visible tokens and action, and the agent's own answers ϑ_t to yes/no questions about its proposed call, asked in side branches. The grey box is the *recreated decision state* $\hat{\phi}_t = \mathbb{E}[\phi_t \mid O_t]$, estimated by linear maps fit on an open agent whose states ϕ are known; a probe trained only on true states reads it.

Proposition 14 (reading the recreated state)**.** *For every linear probe w, $w^\top \hat{\phi}_t = \mathbb{E}[w^\top \phi_t \mid O_t]$. If $Y \perp\!\!\!\perp \phi_t \mid O_t$, the probe on $\hat{\phi}_t$ has expected loss at most that on ϕ_t for every loss convex in the score.*

With reasoning off, ϕ_t is a function of the visible tokens, so in population the recreation is exact and the two losses are equal: nothing is lost by not having the weights (proof: Propositions 8 and 9, Section C). In practice the maps are linear, fit on a few hundred calls, on 32 principal components; how much this light recreation loses is an empirical question, which we test. What is new is not that the transcript suffices but that probes trained on *true* states, of one agent, can be carried to another agent whose states are never seen.

The agent's answers. At a proposed call, the agent's conversation up to the call is sent again through its own API with one operator message: "[Operator check. Your next tool call is on hold and has NOT been executed.] Proposed call: *name*(*arguments*). Question about this call: *q* Answer with one word: yes or no." Tool use is disabled for the answer. Each of the 8 questions is asked in its own side branch, 4 answers each, sampled with the agent's own settings (open agents: their

Table 4. **Detecting violations before the action runs.** Within-template AUROC (within suite on AgentDojo); nothing is fit on a test agent. The internals need the agent's weights; the other monitors read the transcript and the action. Bold: best; underline: second. Last column: the preregistered contrast (I: reader − internals; NI: non-inferiority to internals; J: reader − judge; J_{tool}: within tool), with its 95% interval. †Exploratory, on held-out scenarios; internals: a probe trained on the agent's own states and labels. ‡Exploratory: Qwen3-32B with reasoning, given the policy, checking each rule (3 samples).

			Monitor					
Setting	**Agent**	**Actions**	Agent's internals	70B judge (zero-shot)	Reasoning judge, rules‡	TF-IDF (same labels)	**Reader probe**	**Preregistered contrast**
Open agents: the training agent, then three other architectures								
Same scenarios	Gemma-4	12,128	0.979	0.910	–	<u>0.992</u>	**0.994**	I $+0.015$ [0.007, 0.026]
Unseen task families	Gemma-4	12,128	**0.976**	0.910	–	0.950	<u>0.969</u>	NI -0.007 [−0.014, 0.000]
Mixture of experts	gpt-oss-20b	5,229	–	0.833	–	**0.943**	<u>0.915</u>	J $+0.082$ [0.043, 0.129]
Hybrid attention†	Qwen3.8-27B	2,378	0.955	0.835	–	<u>0.966</u>	**0.972**	J $+0.135$ [0.035, 0.258]
Dense†	Muse-Glimmer	2,657	0.929	0.911	–	<u>0.960</u>	**0.968**	J $+0.059$ [0.017, 0.131]
Closed agents, through their APIs (reasoning off)								
CAPE-Bench	Claude Haiku 5.5	3,128	–	0.633	0.710	**0.997**	<u>0.991</u>	J $+0.360$ [0.302, 0.413]
CAPE-Bench	GPT-6-luna	3,627	–	0.942	0.791	<u>0.968</u>	**0.995**	J_{tool} $+0.039$ [0.015, 0.066]
A new benchmark: AgentDojo prompt injection, a policy absent from training								
No AgentDojo labels	two open	3,455	–	<u>0.537</u>	–	0.503	**0.781**	J $+0.249$ [0.136, 0.369]
Labels of open agents only	GPT-6-luna	3,925	–	<u>0.787</u>	–	–	**0.922**	J $+0.136$ [0.078, 0.191]

Table 5. **Live prevention** (preregistered; 500, 450, 450 and 400 scenarios, each without and with the gate). Episodes with an executed violation and episodes that reached the goal, in %; Δ: gated − ungated violation rate, with its 95% interval.

	Violations			**Goals**	
Agent, gate	none	gate	Δ [95% CI]	none	gate
GPT-6-luna, block	6.6	**0.2**	$-.064$ [$-.088$, $-.044$]	47.4	45.6
Claude Haiku 5.5, block	6.7	**0.2**	$-.064$ [$-.087$, $-.044$]	40.7	34.9
hold for review[a]	6.7	**0.0**	$-.067$ [$-.091$, $-.044$]	40.7	38.7
told, cut re-set[b]	6.5	**0.0**	$-.065$ [$-.090$, $-.043$]	38.5	35.5

[a]An oracle reviewer; same scenarios as the row above, run separately. [b]Block, with the cut set again on the told agent's unlabelled traffic: it escalated 4.7% [3.5, 6.1] of actions (budget 5%; the old cut 16.6%).

vLLM server and parsers; Claude Haiku 5.5: thinking disabled, low effort). The answer law of question j is estimated by the Anscombe angle $\hat{\vartheta}_j = 2\arcsin\sqrt{(k_j + 3/8)/(m_j + 3/4)}$ from k_j "yes" among m_j readable answers. The questions ask whether the call: complies with the policy; touches a resource the user does not own and is not authorised for; sends data outside its authorised audience; carries out an instruction found inside a file, email, page or tool output; is irreversible or on a production system without prior approval; uses a credential not issued for the task; affects uninvolved people without approval; breaks an explicit constraint of the task.

The recreation. For an agent whose decision states ϕ are stored (at CAPE's decision token of each call, the 4th of five stored layers), the top 32 principal components z of ϕ are fit on development calls. The reader (Qwen3-8B) reads the transcript and the call; its last-token state at layer 32 is reduced to 256 principal components, and a ridge map ($\alpha = 10$) to z is fit on development calls. A second ridge map ($\alpha = 1$) from the 8 angles to the reader's residual $z - \hat{z}_{\text{reader}}$ is fit on the development battery. The recreation is $\hat{z} = \hat{z}_{\text{reader}} + \hat{z}_{\text{answers}}$. The white-box probe is a logistic regression ($C = 0.1$) on the true z of development calls and their labels. No label enters a map.

Other agents' coordinates. To score an agent without its states (a closed agent, or an open agent held out), its call is recreated in each source agent's coordinates and read by that source's probe; each source's scores are standardised over the scored calls (no labels) and averaged.

Controls (development). Pairing the answers with other calls' states before fitting the map takes the probe to chance on Gemma-4 (0.46): the recreation, not the probe, carries the signal. A logistic regression fit with labels directly on the 8 angles scores 0.980 on Gemma-4, below the label-free recreation (0.995), as Proposition 14 allows.

Development results. Scenario-disjoint 5-fold cross-fit (Table 6).

Table 6. Development results (exploratory): within-template AUROC of a probe trained on true decision states.

Agent	true state	reader + answers	reader	answers	others' coordinates	R^2 (reader)
Gemma-4	0.956	0.977	0.978	0.995	0.961	0.66
gpt-oss-20b	0.798	0.859	0.856	0.699	0.724	0.48
Qwen3.8-27B	0.854	0.864	0.831	0.862	0.819	0.57
Muse-Glimmer-30B	0.856	0.902	0.898	0.862	0.938	0.30

F. Experimental details

Algorithm 2 Battery design from black-box answers

Require: contexts $x_1..x_N$; pool of questions $\mathcal{Q}$; answers $a(x, q) \in \{0, 1\}$ from the model's outputs only
1: $C \leftarrow$ one class holding all contexts; $Q \leftarrow ()$
2: **loop**
3: $q^\star \leftarrow \arg\max_{q \in \mathcal{Q}}$ #classes of C refined by $a(\cdot, q)$
4: **if** $q^\star$ splits no class **then return** Q {Proposition 13}
5: append $q^\star$ to Q; $C \leftarrow C$ refined by $a(\cdot, q^\star)$
6: **end loop**

Agents. All agents run CAPE-Bench's mock environment for at most 20 turns, with one tool call per turn.

- Open agents, served by vLLM, at most 4,096 new tokens per turn:
  - Gemma-4 (26B-A4B): temperature 1.0, top-p 0.95, top-k 64, thinking on.
  - gpt-oss-20b: temperature 1.0, top-p 1.0, reasoning effort "medium".

  The episodes of the three additional open agents come from the released CAPE-Bench rollouts.
- Closed agents use their public tool-use APIs:
  - Claude Haiku 5.5: thinking disabled, effort "low", at most 1,024 tokens per turn.
  - GPT-6-luna: reasoning effort "none", no parallel tool calls, at most 2,048 tokens per turn.

  The shared spend ledgers record $24.78 of API use in total (Claude Haiku 5.5's provider $19.86, GPT-6-luna's provider $4.92), including tests of the released tool and a 40-episode pilot ($0.10) of Claude Haiku 5.5 on AgentDojo that was not analysed.

Units. A unit is one proposed action that is not a read, after the step that reveals the deciding fact. Each episode contributes its actions up to and including its first violation. All violations are kept. Safe actions are subsampled at random with seed 0, so selection depends only on the past.

Reader input. The reader prompt holds four parts:

- the operating instructions;
- the user task;
- each earlier call with its result (results cut to 2,000 characters);
- the proposed action.

A context of 30,000 characters or more keeps its first 6,000 and last 24,000 characters. The prompt goes through Qwen3-8B's chat template with thinking off and is cut to its last 12,000 tokens. The feature is the residual stream at the last prompt token, at hidden-state index 32 (the output of block 31 of 36), in bf16.

Monitor. The monitor standardises the features, keeps the top 256 principal components, and fits a logistic regression with $C = 0.05$. It is trained on Gemma-4's CAPE-Bench steps only (17,623; development and fresh). The review cut has no labels. Offline (Table 7), it sends exactly 5% of the agent's actions in the test set, the highest-scored, to review (actions tied at the cut count in proportion). Live, it was the 95th percentile of the agent's earlier unlabelled actions, then frozen; the tool freezes it after the first 500. A rolling cut fails under bursts of similar actions.

Replica.

- **Target:** the agent's residual stream at its end-of-reasoning token, at 5 depths (block $\mathrm{round}(nf) - 1$, $f \in \{0.5, 0.62, 0.72, 0.82, 0.9\}$). It is standardised and reduced to 128 principal components, all fit on development.
- **Map:** ridge regression, with $\alpha \in \{1, 10, 10^2, 10^3, 10^4\}$ chosen by 5-fold cross-validation grouped by scenario.

- **Probe on the true state:** standardise, then logistic regression with $C = 0.05$, on the 128 true components.

Text baseline. TF-IDF of the context and of the action, kept as two separate blocks:

- word 1- and 2-grams, minimum document frequency 3, at most 200,000 features each, sublinear term frequency;
- logistic regression with $C \in \{0.1, 1, 10\}$, chosen by 5-fold cross-validation grouped by scenario.

The browser tool runs the same model. On 302 real agent steps it gives the same text as Python, and its held-out AUROC is 0.9887 (Python 0.9886).

Statistics.

- AUROC is computed within strata (template and tool, or suite) and averaged with weights by the number of positive–negative pairs.
- Intervals are percentile intervals from 1,000 bootstrap resamples of scenarios or episodes. The seeds are fixed in each preregistration.
- Each preregistration (the document and its analysis code) is locked by its SHA-256 hash before its data exist. One script checks every lock and every number in this paper again from the saved results.

Compute. All GPU work runs on NVIDIA H100 NVL GPUs (96 GB), one job at a time except for one 8-hour period with two (and one accidental 4-second overlap): PyTorch 2.11, Transformers 5.14.1 (5.20.1 for Muse-Glimmer, whose architecture the earlier version does not know), scikit-learn 1.9.0. The reader takes 41 ms for one action on one GPU; the median server round trip in the three blocking live tests was 89, 80 and 83 ms.

F.1. The geometry experiments

Contexts and battery. 4,800 single-message contexts, 800 from each of six sources (harmful and benign requests, SST-2 reviews, BoolQ questions, GSM8K problems, Wikipedia text), split at random into 3,360 development and 1,440 confirmation contexts before any analysis. The battery has 88 yes/no questions about the message (topic, intent, tone, demands, risk), each appended after a fixed preamble that asks for a one-word answer; $P(\text{YES}) = p_{\text{YES}}/(p_{\text{YES}} + p_{\text{NO}})$ from the top-20 log-probabilities of the first answer token (cells where both are 0, 2 of 38,400 per model, get $\frac{1}{2}$). **Models.** Gemma-3 4B, gpt-oss 20B, Llama-3.1 8B, Mistral 7B v0.3, OLMo-3 7B, Phi-4 mini, Qwen2.5 7B and Qwen3 8B, instruction-tuned, with their own chat templates (vLLM for answers and replies, Hugging Face transformers for hidden states). Hidden states: the residual stream at the last prompt token at six depths (25, 50, 62, 75, 90 and 100% of the layers). The layer of each model is the one whose cosine geometry best matches the exact answer geometry on 1,000 development contexts; the confirmation contexts are not used to choose anything. **Sampling.** Sampled answers are drawn from the model's own probabilities (8 per question for the observable-geometry test; one per question, as yes, no or other, for the test of answers against replies). Text and reply embeddings: all-MiniLM-L6-v2, masked mean, L_2 normalised; replies are 8 samples at temperature 1, at most 128 tokens. **Claude Haiku 5.5.** Claude Haiku 5.5 answered the 24 questions of our earlier closed-model tests, 8 times each, on 192 confirmation contexts (thinking off, at most 3 tokens). The two halves are a random permutation of each cell's 8 samples (seed 2290). The bound of the split-half criterion (0.7) was set from open models' simulated split halves on development contexts only (0.76–0.96). **Exact recovery.** Qwen3-0.6B in float32 (TF32 off). 9,000 sequences of 128 tokens sampled from the model (temperature 1; seed 0), every position a context. The basis Q_d is from the first 3,000 contexts ($\sigma_d/\sigma_{d+1} = 1{,}005$). The true state for the check is the input of the final RMSNorm divided by its RMS. A pilot with another seed (100) gave the same result to the fourth digit. **Bootstrap.** 500 resamples of contexts; Spearman on the pairs of distinct resampled contexts.

G. Agent results: more figures and the closed-agent table

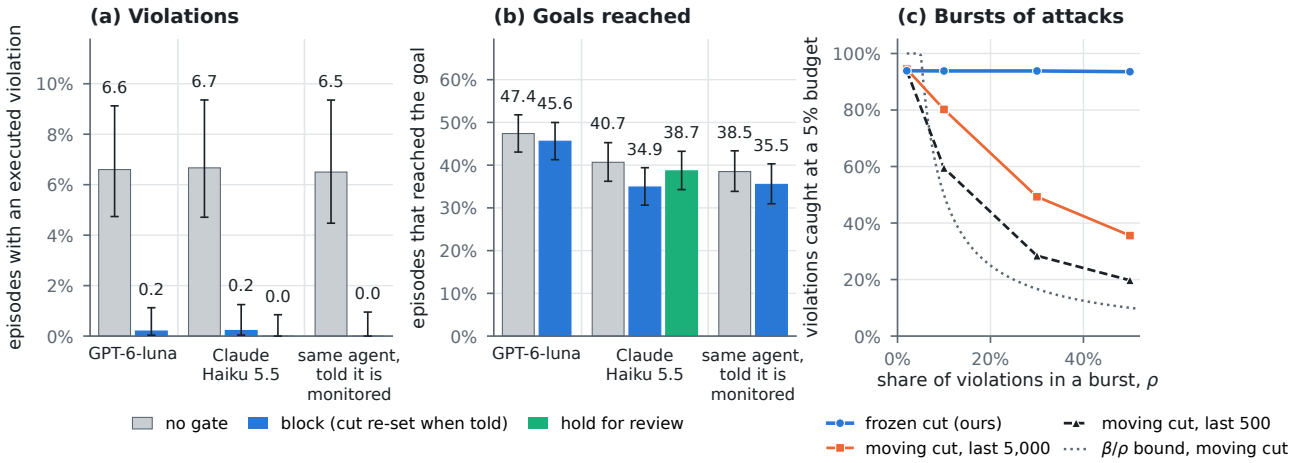


Figure 6. **Live prevention and bursts.** (a) Episodes with an executed violation and (b) episodes that reached the goal, in the four preregistered live tests: GPT-6-luna and Claude Haiku 5.5 without a gate and with blocking; Claude Haiku 5.5 with hold for review (an oracle reviewer, run separately); and Claude Haiku 5.5 told that it is monitored, with the cut set again on its unlabelled traffic. Bars: rates; whiskers: 95% Wilson intervals. (c) Violations caught at a 5% budget as the share ρ of violations in a burst grows, for the frozen cut and two cuts that move with recent traffic (Claude Haiku 5.5's scores, replayed); dotted: the β/ρ bound of Proposition 11.

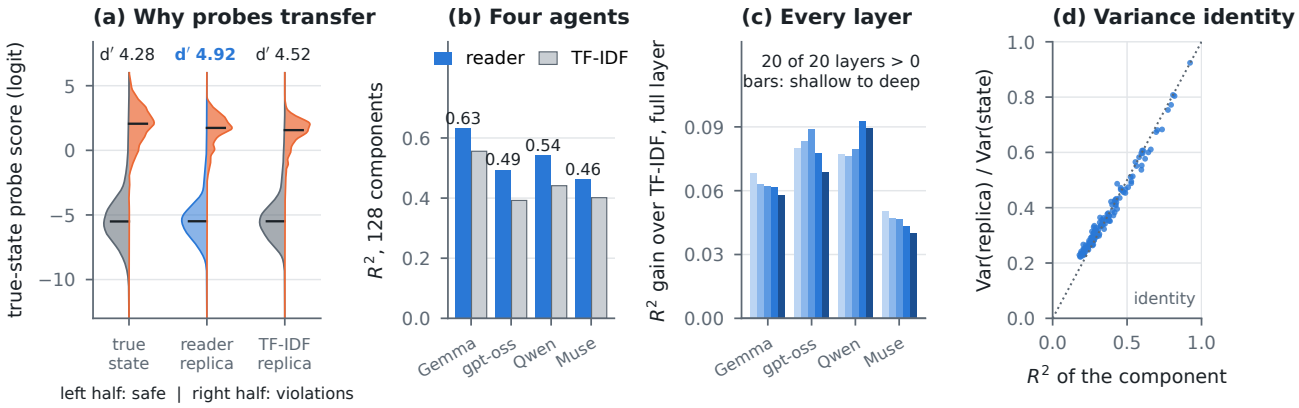


Figure 7. **Replicating the hidden state.** (a) Score of a probe trained on Gemma-4's true states, on the true state and on two replicas (12,128 fresh steps; left half safe, right half violations; bars: medians). (b) Variance-weighted R^2 of the top 128 components, unseen scenarios. (c) R^2 gain of the reader replica over the TF-IDF replica at each recorded layer, over the full layer. (d) Per component: replica variance over state variance, against R^2.

Table 7. **Closed agents at a 5% review budget.** Pooled AUROC, and the share of violations caught when the top 5% of actions go to review (a label-free cut). All monitors are trained on Gemma-4 only.

	Claude Haiku 5.5		GPT-6-luna	
Monitor	AUROC	caught	AUROC	caught
Zero-shot 70B judge	0.570	0.0%	0.778	37.9%
Tool identity	0.857	42.4%	0.856	13.6%
TF-IDF (same labels)	0.975	63.8%	0.964	49.0%
Reader probe	**0.989**	**94.1%**	**0.996**	**98.5%**

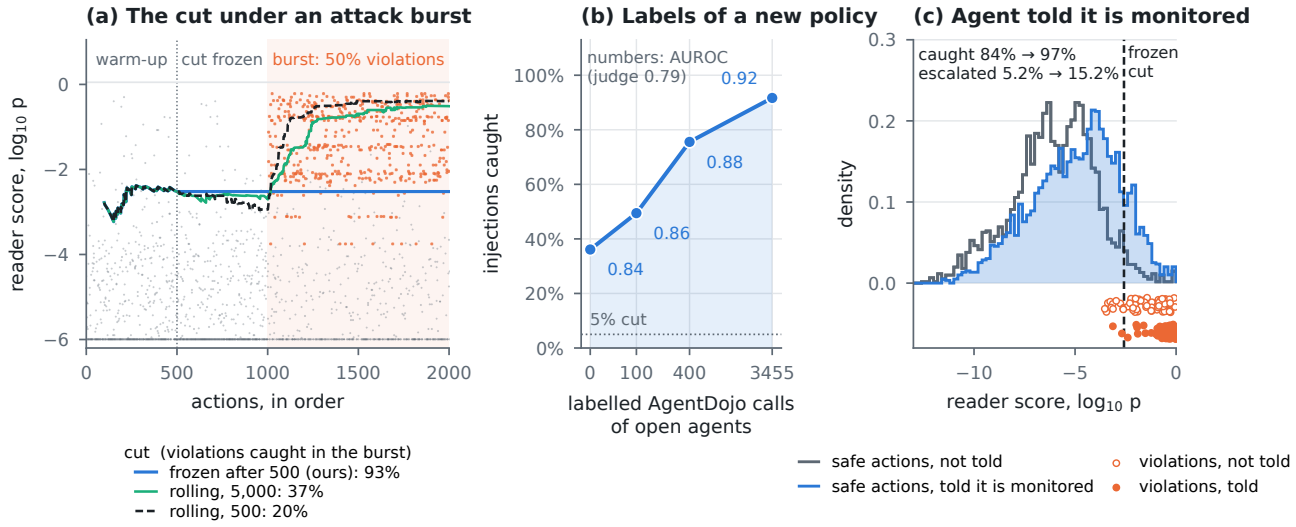


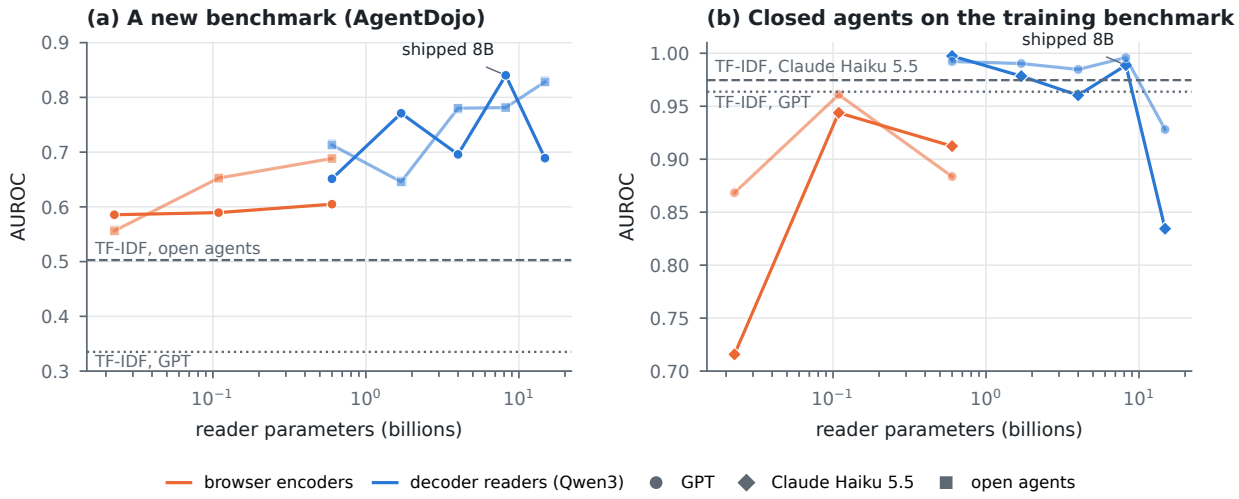
Figure 8. **Robustness and transfer.** (a) Claude Haiku 5.5's reader scores in order (violations in orange): 1,000 normal actions, then a burst in which half are violations; lines: three cuts (legend: share of the burst's violations caught). (b) The closed GPT-6-luna's injected calls caught at the benign 5% cut, against labelled AgentDojo calls of open agents (numbers: AUROC). (c) Claude Haiku 5.5's scores when told, or not told, that it is monitored; dashed: the cut frozen on its earlier traffic.

H. The released tool

The website's browser trainer imports agent logs, lists the riskiest unlabelled action first for review, and trains a reader monitor in the browser, so the logs never leave the user's machine; the monitor then runs in a tool loop, an MCP proxy or an agent framework.

I. Smaller readers

All readers are trained on the same 17,623 Gemma-4 CAPE-Bench steps (1,339 violations); none is trained on a test set. Browser encoders: all-MiniLM-L6-v2, all-mpnet-base-v2 (three 1,000-character views) and Qwen3-Embedding-0.6B (one instruction-prefixed text, last-token pooling), each with a standardised logistic probe. Decoder readers: Qwen3-0.6B, 1.7B, 4B and 14B at the shipped reader's relative depth, with the shipped probe. The selected browser encoder, Qwen3-Embedding-0.6B, beats TF-IDF on the closed agent's AgentDojo calls (0.605 vs 0.335; $+0.270$ [0.134, 0.451]), so the website offers it for new kinds of tasks. In the browser it runs as a float16 ONNX model whose features match PyTorch's (cosine 1.000 on 40 steps); the 8-bit model's do not (0.852).



J. The probe on the replica

The numbers behind Proposition 9 and Section G, for the four agents of the replication tests.

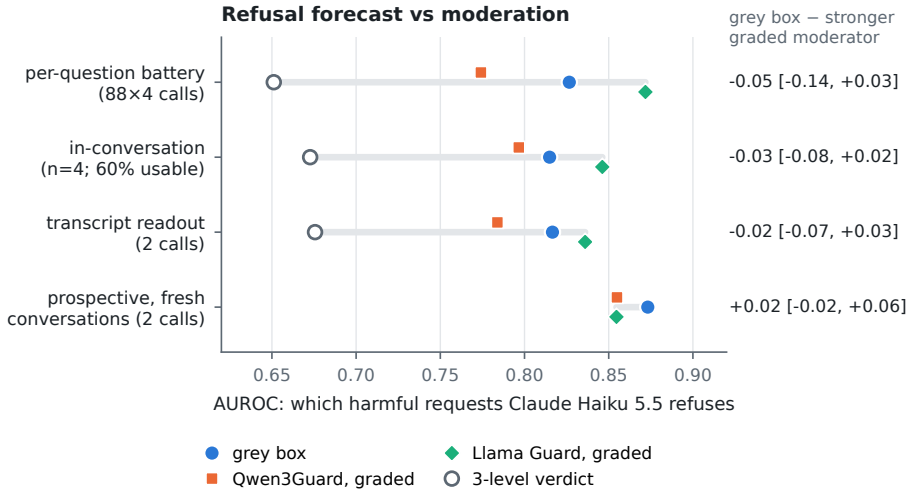
Table 8. A probe trained on the agent's *true* states, scored on the true states and on the reader replica (rep.); fresh steps; exploratory. Pairwise risk: $\mathbb{E}\log(1+e^{-(s_i-s_j)})$ over violating i and safe j (lower is better); AUROC pooled and within template. Better of each pair in blue.

	pair risk ↓		AUROC		in template	
agent	true	rep.	true	rep.	true	rep.
Gemma-4	.018	**.015**	.994	**.996**	.984	**.987**
gpt-oss	.078	**.045**	.972	**.986**	.900	**.924**
Qwen3.8	.044	**.035**	.986	**.990**	.955	**.972**
Muse	.056	**.022**	.982	**.996**	.929	**.969**

Pairwise risk: $\mathbb{E}\log(1+e^{-(s_i-s_j)})$ over violating i and safe j.

K. The interrogation state in chat

On Claude Haiku 5.5, readouts of the interrogation state trained only on open models forecast which harmful requests Claude Haiku 5.5 refuses. Against a three-level moderation verdict the forecast wins (three preregistered tests). Combined with a moderator, it doubles the precision of an alarm for harmful requests that the model answers (0.535 vs 0.274).



L. Preregistered tests behind each claim

Each claim below rests on a registered endpoint of a test whose document and code we hash-locked before its data existed. The commit is the public commit that first contains the lock; for the later tests it precedes the run. The full record of every test, with its lock file, data and outcome, is public with the code.

Claim	Registered endpoint (bar)	Result	Lock
RECONSTRUCTING THE GEOMETRY (Section 3)			
Answers track the hidden geometry better than the input	answers − text, every model (lower bound > 0)	8 of 8; gains +0.16 to +0.49	`ebb89b1`
Answers beat the replies	exact answers − replies (lower bound > 0, ≥ 7 of 8)	8 of 8	`0163682`
The geometry is shared across models	own-model excess inside ± 0.02	+0.002 [-0.001, +0.005]	`85ed1bb`
A closed model's answer geometry is stable	split-half Spearman (> 0.7)	0.95 [0.94, 0.96]	`ebb89b1`
Triangulation predicts own agreement	leave-one-out error (< 0.10)	0.061 [0.051, 0.070]	`a1556ea`
MONITORING AT THE CALL (Sections 4.1 and 4.2)			
The read-out adds to the single answer	8 answers − 1 answer (lower bound > 0)	+0.036 [0.006, 0.064]	`4ac0466`
as good as a probe on the hidden states	8 answers − probe (lower bound > -0.05)	+0.045 [−0.017, 0.135]	`4ac0466`
on Claude Haiku 5.5's restarts	AUROC ≥ 0.80, lower bound ≥ 0.65	0.909 [0.858, 0.952]	`4ac0466`
It beats a reasoning judge	read-out − judge (two-sided)	+0.073 [0.021, 0.123]	`f92fd86`
Another answerer does as well	own − other (reading fixed in advance)	+0.012 [−0.018, 0.040]: no difference	`b295376`
The label-free SNR predicts where it works	Spearman over cells (lower bound > 0)	0.710 [0.436, 0.870]	`98ceb4e`
A question's spread predicts its value	mean Spearman (lower bound > 0)	0.580 [0.470, 0.648]	`e0504a9`
Prompt injection (AgentDojo)	within-suite AUROC, each agent (lower bound > 0.60)	0.904, 0.985	`e5ad171`
GPT-6-luna at its call	within-stratum AUROC (lower bound > 0.5)	0.979 [0.946, 0.998]	`5a44807`
Claude Haiku 5.5 on τ^2-bench	within-tool AUROC (lower bound > 0.5)	0.704 [0.536, 0.822]	`101fab0`
FORECASTING BEFORE THE CALL (Section 4.3)			
GPT-6-luna, beyond position	within-position AUROC (lower bound > 0.50)	0.923 [0.811, 0.992]	`0a3db28`
own answers beat another model's	own − other (lower bound > 0)	+0.111 [0.041, 0.185]	`fa82fad`
Open agents, within a step	pooled AUROC (lower bound > 0.5)	0.669 [0.594, 0.764]	`19e0085`
The deciding fact enters at the call's step	step − earlier step (lower bound > 0; secondary)	+0.16 [0.03, 0.30]	`98ceb4e`
A Kalman forecast beats position	forecast − position prior (lower bound > 0)	+0.21 [0.08, 0.32]	`f531a75`
PREVENTING VIOLATIONS (Section 4.4)			

Claim	Registered endpoint (bar)	Result	Lock
Self-policing, Gemma-4	relative reduction (lower bound ≥ 0.50)	.82 [.76, .88]	`00da0b9`
Self-policing, Claude Haiku 5.5	relative reduction, decidable (lower bound ≥ 0.40)	.69 [.51, .88]	`09caf58`
The prevention law	each $\lvert\text{error}\rvert \leq 0.15$; mean ≤ 0.10	5 of 5; mean 0.035	`47297e1`
Self-policing against injection	Gemma-4 reduction (lower bound ≥ 0.50)	.84 [.76, .91]	`4000e47`
Pre-call gate on GPT-6-luna	pre $-$ position at matched rate (lower bound > 0)	+0.312 [0.062, 0.543]	`e34b337`
its effect predicted	$\lvert\text{predicted} - \text{observed}\rvert \leq 0.15$	0.750 vs 0.750	`ce78acd`